,,,

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Highlights

A Survey on Multi-Robot Collaboration and Communication: Taxonomy, Challenges, and Future Directions

Owais Ahmed, Adeel Khalid, Daniel Byers, M. Hassan Tanveer

- Comprehensive taxonomy of multi-robot collaboration spanning task allocation, communication frameworks, and learning paradigms
- Critical analysis of distributed SLAM, opportunistic communication, and LLM-driven coordination advances
- Comparative evaluation of 50+ representative systems across scalability, communication efficiency, and real-world validation
- Identification of open challenges and future directions for scalable, interpretable, and trustworthy multi-robot systems

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

A Survey on Multi-Robot Collaboration and Communication: Taxonomy, Challenges, and Future Directions

Owais Ahmed[a,*], Adeel Khalid[a], Daniel Byers[b] and M. Hassan Tanveer[b]

[a]College of Engineering and Engineering Technology, Kennesaw State University, Kennesaw, Georgia, 30144, USA
[b]Department of Robotics and Mechatronics Engineering, Kennesaw State University, Kennesaw, Georgia, 30144, USA

ARTICLE INFO

Keywords:
Multi-Robot Systems
Multi-agent coordination
Collaborative SLAM
Distributed learning
Communication protocols
Human-robot interaction
Task allocation
Large Language Models

ABSTRACT

Multi-robot collaboration and communication enable teams of autonomous agents to collectively sense, plan, and act in applications spanning search and rescue, precision agriculture, warehouse logistics, and disaster response. Despite significant algorithmic progress, coordinating large heterogeneous robot teams in bandwidth-constrained, dynamic environments remains challenging. This survey consolidates state-of-the-art advances across communication architectures, task allocation strategies, perception frameworks, and learning paradigms. We present a comprehensive taxonomy organized into three primary dimensions: task allocation and coordination (including Large Language Model (LLM)-based planning and human-in-the-loop interfaces), communication and mapping frameworks (encompassing distributed Simultaneous Localization and Mapping (SLAM), opportunistic protocols, and collaborative perception), and learning and control methods (spanning multi-modal deep reinforcement learning and distributed optimization). Particular emphasis is placed on emerging paradigms including spectral prioritization for bandwidth-efficient SLAM, opportunistic communication for intermittent connectivity, semantic and mechanical interaction, protocol adaptability for agricultural and warehouse applications, and reinforcement learning for dynamic task allocation with infeasible task handling. Our comparative analysis evaluates scalability, communication efficiency, interpretability, and real-world validation across representative frameworks. We identify critical open challenges and outline future research directions toward scalable, interpretable, and trustworthy multi-robot systems deployable in real-world environments.

1. Introduction

Multi-robot collaboration enables teams of autonomous agents to collectively sense, plan, and act in environments that are too complex or hazardous for a single robot. Such systems are increasingly deployed in domains such as warehouse logistics, precision agriculture, search and rescue, and planetary exploration. By distributing sensing and computation, Multi-Robot Systems (MRS) offer scalability, redundancy, and robustness against individual failures.

Despite their promise, coordinating large robot teams in real-world settings remains formidable. Challenges arise from bandwidth-constrained communication, heterogeneous hardware capabilities, dynamic and unstructured environments, and the need for shared situational awareness. Traditional MRS frameworks have relied on centralized task allocation or decentralized consensus-based control, both of which struggle to balance optimality, fault tolerance, and scalability under uncertain conditions.

Recent advances in Artificial Intelligence (AI) (particularly Large Language Models (LLMs), Vision-Language Models (VLMs), and Graph Neural Networks (GNNs)) have redefined how multi-robot systems communicate, reason, and collaborate. LLM-driven agents can interpret natural language commands, decompose high-level goals into executable subtasks, and reason about robot capabilities in human-interpretable ways. VLMs provide semantic scene understanding that goes beyond geometric perception, while GNNs enable distributed perception and decision-making through structured message passing. Together, these technologies are steering multi-robot collaboration toward explainable, communication-efficient, and human-aware systems.

Recent work has further highlighted the importance of agent-server architectures that balance onboard computation with centralized processing [5], game-theoretic approaches for social navigation in constrained spaces [10], and neural network-based methods for coordinated exploration [58]. Distributed relative localization using only range measurements [61] and optimization-based approaches for task allocation under communication constraints [49] demonstrate the field's evolution toward practical, constraint-aware systems. Heterogeneous teams combining aerial and ground robots for environmental monitoring [28] and privacy-preserving coordination using homomorphic encryption [29] address emerging requirements for security and cross-domain collaboration.

Recent distributed Simultaneous Localization and Mapping (SLAM) frameworks have demonstrated remarkable progress in enabling large robot teams to collaboratively map environments without centralized coordination. Swarm-SLAM [30] introduces spectral prioritization for loop closure selection, achieving near-centralized mapping quality while reducing communication overhead by 60%. SlideSLAM [35]

*Corresponding author
oahmed7@students.kennesaw.edu (O. Ahmed); akhalid2@kennesaw.edu (A. Khalid); dbyers10@students.kennesaw.edu (D. Byers); mtanveer@kennesaw.edu (M.H. Tanveer)
ORCID(s): 0009-0008-2600-6650 (O. Ahmed); 0009-0001-3322-6677 (A. Khalid); 0009-0002-2960-6332 (D. Byers); 0000-0001-9266-6368 (M.H. Tanveer)

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

extends these principles to heterogeneous teams with metric-semantic representations, enabling robust place recognition across different sensor modalities. Distributed variational inference methods further advance object-level SLAM by enabling probabilistic pose estimation across robot teams [9]. Localization and Mapping Platform (LAMP) 2.0 [11], validated in Defense Advanced Research Projects Agency (DARPA) Subterranean Challenge environments, demonstrates resilient multi-robot SLAM under extreme conditions including Global Positioning System (GPS) denial, communication degradation, and perceptual aliasing in underground mines. While object-level SLAM is fundamentally a perception and mapping problem, such representations directly inform coordination and task allocation by enabling robots to reason over shared semantic entities rather than low-level geometry.

Opportunistic communication strategies address scenarios where continuous connectivity cannot be guaranteed. Miller et al. [39] developed Multi-robot Opportunistic Communication for Heterogeneous Collaboration (MOCHA), enabling large-scale heterogeneous collaboration through gossip-based information exchange that tolerates intermittent connectivity while maintaining global situational awareness across aerial and ground platforms.

Yet, fundamental questions persist about how robots should communicate (not merely through digital packets, but through diverse modalities that include physical interaction, semantic intent sharing, and protocol selection adapted to environmental constraints). Human operators increasingly participate in the coordination loop, necessitating interfaces that support both direct verbal interaction and indirect communication through shared visual representations. Meanwhile, real-world deployments in agriculture, warehouses, and disaster zones expose the brittleness of communication assumptions, demanding protocols that adapt to range limitations, signal degradation, and intermittent connectivity.

This survey consolidates recent progress in multi-robot collaboration and communication, highlighting the transition from classical coordination methods to modern learning-based and language-driven frameworks. The paper presents a unified taxonomy encompassing communication architectures (from concurrent transmission mechanisms to semantic and mechanical interaction), coordination strategies that span centralized, decentralized, and hybrid paradigms, and learning frameworks that include distributed neural network optimization and multi-modal perception fusion. We examine representative applications across Unmanned Aerial Vehicle (UAV) swarms, warehouse robotics, agricultural automation, and household assistance, critically analyzing their scalability, interpretability, and real-world validation. Finally, we identify open challenges and outline future research directions for scalable and trustworthy multi-robot cooperation.

The remainder of this paper is organized as follows: Section 2 describes the survey methodology and scope. Section 3 presents communication fundamentals for multi-robot systems. Section 4 reviews foundational and recent work on multi-robot collaboration. Section 5 summarizes key communication and coordination architectures including concurrent transmission, protocol analysis, and multi-modal interaction. Section 6 discusses representative applications across diverse domains. Section 7 provides a comparative analysis of scalability, interpretability, and real-world validation, followed by open challenges and future research directions in Sections 8 and 9, respectively.

2. Survey Methodology and Scope

This survey examines communication architectures and coordination mechanisms for multi-robot collaboration, covering literature from 2019 to 2025 with emphasis on recent advances. We focus on systems where communication plays a central role in enabling coordination, encompassing both algorithmic innovations and practical deployment considerations.

2.1. Scope and Selection Criteria

The scope of this survey encompasses multi-robot systems where two or more autonomous or semi-autonomous robots communicate to achieve shared objectives. We include both homogeneous teams (identical robots) and heterogeneous teams (robots with different capabilities, sensors, or platforms). The platforms considered range from ground vehicles and aerial drones to legged robots and manipulators.

Papers were selected based on the following criteria: (1) the work addresses communication aspects of multi-robot systems, including protocol design, information exchange strategies, or communication-aware coordination; (2) the contribution is validated through simulation, real-robot experiments, or mathematical analysis; (3) the work appears in peer-reviewed venues including IEEE, ACM, Springer, and Elsevier publications. We prioritize works with real-world validation or those introducing novel communication paradigms.

2.2. Taxonomy Overview

Our taxonomy organizes multi-robot collaboration and communication along three primary dimensions, as illustrated in Fig. 1:

Task Allocation and Coordination: Methods for assigning tasks to robots and coordinating their activities, including LLM-based planning (RoCo, COHERENT, HMAS), human-in-the-loop semantic coordination (HMCF), auction and market-based allocation, Swarm-SLAM and GNN-based mapping, GNN-based goal assignment (DGNN-GA), collaborative semantic mapping, multi-human multi-robot interaction, agent-server coordination, task allocation with range constraints, heterogeneous team coordination, temporal logic task planning, and collaborative object-level mapping.

Communication, Mapping, and Physical Coordination Frameworks: The infrastructure and protocols enabling information exchange, including wireless and ROS-based frameworks, decentralized and hybrid coordination, communication-aware coordination (ACHORD), task-driven map compression, distributed and federated learning (DiNNO), mechanical and force-based communication

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

(Omnid Mocobots), concurrent transmission architectures, protocol-aware communication (Wi-Fi, Bluetooth, Zigbee, LoRa), collaborative perception with centralized fusion, multi-UAV topological exploration, distributed object-level SLAM, semantic-relational collaborative SLAM (Multi S-Graphs), sparse and swarm-scale SLAM, and decentralized pose-graph optimization.

Learning Paradigms and Control: Data-driven methods for adaptive coordination and control, including ability-aware control (DA3C), multi-modal deep reinforcement learning, language-driven learning, communication-efficient collaborative learning (CoCoL), heterogeneous multi-agent reinforcement learning (HetGPPO), UAV 3D occupancy prediction, hierarchical macro-action learning, GNN-based submodular action selection, learning communication strategies (MARL), MARL for resource allocation, collaborative dense reconstruction, optimal coverage path planning, and learning-assisted SLAM coordination.

3. Communication Fundamentals for Multi-Robot Systems

Effective multi-robot collaboration requires understanding the communication constraints that shape system design. This section presents the fundamental concepts, models, and metrics that characterize multi-robot communication.

3.1. Channel Characteristics and Propagation Models

Wireless communication between robots is subject to path loss, multipath fading, and interference that vary with environment and mobility. The received signal power P_r at distance d from a transmitter with power P_t follows the log-distance path loss model:

$$P_r(d) = P_t - PL(d_0) - 10n \log_{10}\left(\frac{d}{d_0}\right) + X_\sigma \quad (1)$$

where $PL(d_0)$ is the path loss at reference distance d_0, n is the path loss exponent (typically 2 for free space, 2.7 to 3.5 for urban environments, and 4 to 6 for indoor or obstructed settings), and X_σ represents log-normal shadowing with standard deviation σ (typically 4 to 12 dB).

For multi-robot systems, the path loss exponent and shadowing characteristics depend heavily on the deployment environment. Underground tunnels exhibit severe multipath and high path loss exponents ($n > 4$), while open agricultural fields approach free-space conditions ($n \approx 2$). Warehouse environments with metal shelving introduce significant shadowing and reflections.

The Signal-to-Noise Ratio (SNR) determines achievable data rates and error probabilities:

$$\text{SNR} = \frac{P_r}{N_0 B} \quad (2)$$

where N_0 is the noise power spectral density and B is the channel bandwidth. The Shannon capacity provides an upper bound on achievable data rate:

$$C = B \log_2(1 + \text{SNR}) \quad (3)$$

For mobile robots, Doppler shift from relative motion introduces additional challenges, particularly for high-speed aerial platforms where velocities can exceed 20 m/s.

3.2. Network Topology and Connectivity

Multi-robot networks can be modeled as graphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where vertices $\mathcal{V} = \{1, 2, \ldots, N\}$ represent robots and edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represent communication links. An edge $(i, j) \in \mathcal{E}$ exists if robots i and j can communicate directly, typically determined by distance and SNR thresholds.

The adjacency matrix $A \in \{0, 1\}^{N \times N}$ has $A_{ij} = 1$ if $(i, j) \in \mathcal{E}$ and $A_{ij} = 0$ otherwise. The degree matrix $D = \text{diag}(d_1, \ldots, d_N)$ contains node degrees $d_i = \sum_j A_{ij}$. The graph Laplacian is defined as:

$$L = D - A \quad (4)$$

The eigenvalues of L, denoted $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$, characterize network connectivity. The second-smallest eigenvalue λ_2, known as the algebraic connectivity or Fiedler value, measures how well-connected the network is. A network is connected (every robot can reach every other robot through multi-hop paths) if and only if $\lambda_2 > 0$. Larger values of λ_2 indicate more robust connectivity that can tolerate link failures.

This spectral characterization is particularly important for distributed SLAM, where Swarm-SLAM [30] uses the Laplacian eigenvalues to prioritize loop closures that maximally increase algebraic connectivity, thereby improving map consistency with minimal communication.

3.3. Communication Architectures

Multi-robot communication architectures fall into three categories:

Centralized Architecture: All robots communicate through a central coordinator (base station or designated leader). This architecture simplifies coordination and enables globally optimal decisions but creates a single point of failure and communication bottleneck. The central node must handle $O(N)$ message traffic, limiting scalability.

Decentralized Architecture: Robots communicate directly with neighbors without central coordination. Each robot maintains local state and achieves global objectives through distributed algorithms (e.g., consensus protocols). This architecture provides fault tolerance and scales better but requires careful algorithm design to ensure convergence and consistency.

Hybrid Architecture: Combines aspects of both paradigms. For example, a central planner may provide high-level task assignments while robots coordinate locally for collision avoidance and detailed execution. Agent-server architectures [5] offload computationally intensive tasks to a central server while maintaining lightweight autonomy onboard each robot.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

3.4. Key Communication Metrics

We define the following metrics to characterize and compare multi-robot communication systems:

Bandwidth Efficiency: The ratio of task-relevant information transmitted to total data volume. Systems that share raw sensor data have low efficiency, while those using compressed features or semantic representations achieve higher efficiency. For distributed SLAM, this is often measured as the number of loop closures or map constraints shared relative to total candidates. Task-driven map compression approaches [42] further maximize bandwidth efficiency by transmitting only compressed map representations tailored to the downstream task (e.g., path planning), achieving up to 98% data reduction while preserving path quality.

Communication Latency: The delay between information generation at one robot and its availability at another. This includes transmission time, propagation delay, and processing overhead. For time-critical coordination (e.g., collision avoidance), latency requirements may be on the order of milliseconds.

Connectivity Robustness: The ability to maintain coordination under link failures or network partitions. Measured by the minimum number of link failures that disconnect the network, or by performance degradation under specified packet loss rates.

Information Freshness (Age of Information): The time elapsed since the most recent update was generated. For dynamic environments, stale information can lead to coordination failures. The Age of Information (AoI) at time t for information generated at time u is $\Delta(t) = t - u$.

Scalability: How communication overhead grows with team size. Centralized systems typically exhibit $O(N)$ message complexity at the coordinator, while well-designed decentralized systems can achieve $O(1)$ per-robot complexity.

3.5. Communication-Constrained Coordination

Many coordination problems must explicitly account for communication constraints. For example, task allocation under connectivity constraints can be formulated as:

$$\begin{aligned} \min_{x} \quad & \sum_{i,j} c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i} x_{ij} = 1, \quad \forall j \quad \text{(task assignment)} \\ & \lambda_2(\mathcal{G}(x)) \geq \epsilon \quad \text{(connectivity)} \end{aligned} \tag{5}$$

where $x_{ij} \in \{0, 1\}$ indicates whether robot i is assigned to task j, c_{ij} is the assignment cost, $\mathcal{G}(x)$ is the communication graph induced by the assignment, and $\epsilon > 0$ ensures network connectivity is maintained. Such formulations enable principled trade-offs between task performance and communication requirements.

3.6. Spectral Methods for Communication-Efficient SLAM

The pose graph in collaborative SLAM can be formally defined as [30]:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}^{\text{local}}, \mathcal{E}^{\text{global}}) \tag{6}$$

$$\begin{aligned} \mathcal{V} &= (\mathcal{V}_1, \dots, \mathcal{V}_n) \\ \mathcal{E}^{\text{local}} &= (\mathcal{E}_1^{\text{local}}, \dots, \mathcal{E}_n^{\text{local}}) \\ \mathcal{E}^{\text{global}} &= (\mathcal{E}_{\text{fixed}}^{\text{global}}, \mathcal{E}_{\text{candidate}}^{\text{global}}) \end{aligned}$$

where $\mathcal{V}$ contains vertices from all n robots' pose graphs (each vertex corresponding to a keyframe), $\mathcal{E}^{\text{local}}$ contains local edges such as odometry measurements and intra-robot loop closures, and $\mathcal{E}^{\text{global}}$ contains inter-robot loop closures divided between fixed measurements already computed and candidate loop closures for prioritization.

The algebraic connectivity of the pose graph controls the worst-case estimation error. The rotation-weighted Laplacian has entries for vertex pairs (i, j) defined as [30]:

$$L_{ij} = \begin{cases} \sum_{(i,j') \in \delta(i)} \kappa_{ij'}, & i = j \\ -\kappa_{ij}, & \{i, j\} \in \mathcal{E} \\ 0, & \{i, j\} \notin \mathcal{E} \end{cases} \tag{7}$$

where κ_{ij} denotes the edge weight and $\delta(i)$ is the set of edges incident to vertex i. Using the similarity score $s_e \in [0, 1]$ from global matching as confidence metric, the weights are defined as $\kappa_{ij} = 1$ for all $e \in (\mathcal{E}^{\text{local}}, \mathcal{E}_{\text{fixed}}^{\text{global}})$ and $\kappa_{ij} = s_e$ for all $e \in \mathcal{E}_{\text{candidate}}^{\text{global}}$.

The Laplacian can be expressed as the sum of subgraph Laplacians:

$$L(\omega) \triangleq L_{\mathcal{E}^{\text{local}}} + L_{\mathcal{E}_{\text{fixed}}^{\text{global}}} + \sum_{e \in \mathcal{E}_{\text{candidate}}^{\text{global}}} \omega_e L_e \tag{8}$$

where $\omega_e \in \{0, 1\}$ determines prioritization of candidate edge e. The loop closure prioritization problem via algebraic connectivity maximization is formulated as [30]:

$$\begin{aligned} \max_{\omega_e \in \{0,1\}} \quad & \lambda_2(L(\omega)) \\ \text{s.t.} \quad & |\omega| = B \end{aligned} \tag{9}$$

where B is the communication budget. This problem is NP-Hard due to the integrality constraint, so it is solved via convex relaxation.

3.7. Distributed Learning Formulations

For distributed neural network optimization, consider N robots operating in a connected communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where robot i has local data $\mathcal{D}_i$. The distributed learning problem is formulated as [60]:

$$\min_{\theta \in \mathbb{R}^d} \quad \sum_{i \in \mathcal{V}} \ell(\theta; \mathcal{D}_i) \tag{10}$$

where $\ell(\cdot)$ is the loss function and θ represents the neural network parameters. Separating the decision variable such that each robot maintains its own instance θ_i yields:

$$\begin{aligned} \min_{\theta} \quad & \sum_{i \in \mathcal{V}} \ell(\theta_i; \mathcal{D}_i) \\ \text{s.t.} \quad & \theta_i = \theta_j \quad \forall (i, j) \in \mathcal{E} \end{aligned} \tag{11}$$

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

The DiNNO algorithm [60] solves this using consensus ADMM with an augmented Lagrangian:

$$\mathcal{L}_a = \sum_{i\in\mathcal{V}} \ell(\theta_i) + p_i^\top \theta_i + \frac{\rho}{2} \sum_{j\in\mathcal{N}_i} \|\theta_i - z_{ij}\|_2^2 \tag{12}$$

where p_i represents the dual variable enforcing agreement between robot i and its neighbors $\mathcal{N}_i$, and ρ is the penalty parameter. The distributed update equations are:

$$p_i^{k+1} = p_i^k + \rho \sum_{j\in\mathcal{N}_i} (\theta_i^k - \theta_j^k) \tag{13}$$

$$\theta_i^{k+1} = \arg\min_\theta \ \ell(\theta; \mathcal{D}_i) + \theta^\top p_i^{k+1} + \rho \sum_{j\in\mathcal{N}_i} \left\| \theta - \frac{\theta_i^k + \theta_j^k}{2} \right\|_2^2 \tag{14}$$

For convex objective functions with L-smooth and strongly convex local objectives, DiNNO converges to the unique global solution with linear convergence rate [60]. In practice, the primal update (14) is approximated using a small number of stochastic gradient steps, enabling application to nonconvex deep learning objectives.

4. Related Work

Multi-Robot Systems (MRS) distribute sensing, computation, and actuation across teams of robots, providing parallelism, resilience, and scalability. They have been deployed in mission-critical settings such as disaster response, warehouse automation, and planetary exploration [43, 12]. However, enabling such teams to coordinate effectively in unstructured, communication-constrained environments remains an open challenge.

Classical approaches to multi-robot coordination rely on centralized or decentralized architectures. Centralized systems achieve globally optimal task allocations but face single points of failure and poor scalability. In contrast, decentralized and consensus-based methods improve robustness and autonomy but often require continuous communication to maintain shared situational awareness. Hybrid models attempt to balance both paradigms by combining global planning with local decision-making, offering scalability and fault tolerance in dynamic environments.

Recent breakthroughs in artificial intelligence have significantly reshaped the MRS landscape. Large Language Model (LLM)-based planners allow robots to reason in natural language, negotiate tasks, and incorporate human feedback through interpretable dialogue [37, 13]. Vision-Language Models (VLMs) extend this capability by introducing semantic understanding into exploration and navigation tasks, enabling robots to infer spatial and contextual cues, such as object location or scene intent [1]. Distributed learning frameworks like **DiNNO** achieve centralized-level performance without aggregating raw data, thereby reducing bandwidth usage and preserving privacy across large robot teams [60].

In addition to algorithmic advances, system-level frameworks have demonstrated the practical feasibility of real-world multi-robot collaboration. Studies on wireless communication networks [24], ROS-based disaster exploration systems [57], and cooperative aerial-ground construction frameworks [27] highlight progress in integrating perception, control, and coordination across heterogeneous platforms. These systems provide valuable insights into how real-time communication and distributed control can be achieved under physical and bandwidth constraints.

Recent contributions such as **ACHORD** and **CoCoL** further advance communication-efficient collaboration. ACHORD introduces communication-aware coordination, allowing robots to maintain connectivity through adaptive relay deployment and prioritized message transmission in intermittent networks. CoCoL focuses on decentralized collaborative learning, achieving high model accuracy while significantly reducing bandwidth consumption through efficient parameter synchronization across heterogeneous teams. Together, these frameworks underscore the increasing focus on optimizing both information exchange and learning efficiency in large-scale MRS deployments.

Human-centered collaboration has also emerged as a key research direction. The **HMCF** [32] integrates human oversight into LLM-driven coordination, where each robot operates with its own LLM agent for task verification while a central assistant LLM manages global planning and communication. This framework demonstrated superior task success rates and adaptability in both simulation and real-world experiments, highlighting the importance of human feedback in ensuring safety and interpretability.

Parallel to cognitive coordination, physical collaboration frameworks like the **Omnid Mocobots** [17] emphasize intuitive and safe interaction between humans and robots. The system uses mechanical communication where robots coordinate through shared forces acting on the payload instead of wireless communication to achieve seamless cooperative manipulation. Experiments with multiple mocobots and human operators validated its ability to handle heavy and articulated payloads naturally, showcasing a complementary dimension of collaboration that relies on tactile and force-based cues rather than data exchange.

Communication forms the backbone of any collaborative multi-robot system. Traditional asynchronous transmission strategies suffer from collision-induced delays and high energy consumption, limiting their applicability in time-constrained scenarios. Bharadwaj et al. introduced **concurrent transmission** as a faster alternative for multi-robot coordination [7]. Their split architecture separates the Networking Unit (NU) from the Computing and Control Unit (CCU), enabling microsecond-level synchronization required for concurrent transmission while maintaining flexibility for heterogeneous robot platforms. Using the Glossy protocol for one-to-all dissemination in a leader-follower setting, the system achieved centimeter-level spatial accuracy and millisecond-level temporal coordination over multi-hop networks with low radio duty-cycling. Real-world experiments

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

with five differential-drive robots demonstrated formation control and dynamic speed variation with near-real-time performance, opening pathways for scalable, energy-efficient multi-robot communication.

Multi-robot SLAM has evolved from centralized architectures toward fully decentralized frameworks that eliminate single points of failure while maintaining map consistency. Lajoie et al. introduced **Swarm-SLAM** [30], a sparse decentralized collaborative SLAM system that uses spectral prioritization to select the most informative inter-robot loop closures. By analyzing the eigenvalues of the pose graph Laplacian, Swarm-SLAM identifies loop closures that maximally reduce map uncertainty while transmitting up to 60% fewer constraints compared to naive sharing strategies. This spectral approach ensures near-centralized mapping quality even in bandwidth-constrained scenarios, demonstrated through extensive validation in GPS-denied environments.

Building on these distributed SLAM principles, Liu et al. developed **SlideSLAM** [35], which extends collaborative mapping to heterogeneous robot teams using metric-semantic representations. Unlike geometric feature-based systems, SlideSLAM leverages object-level landmarks detected from RGB-D cameras or LiDARs, providing viewpoint-invariant place recognition across different sensing modalities. The system supports real-time decentralized operation onboard resource-constrained platforms, with semantics-driven algorithms that match object configurations rather than low-level features. Validated across aerial UAVs, ground UGVs, and legged robots in both indoor and outdoor environments, SlideSLAM demonstrates the advantages of semantic abstraction for heterogeneous multi-robot coordination.

Cao et al. addressed uncertainty quantification in multi-robot object SLAM through distributed variational inference [9]. Their approach enables robots to collaboratively estimate object poses while maintaining Bayesian uncertainty estimates across the team, providing probabilistic guarantees on map consistency without centralized aggregation. The method scales to large teams through message-passing on factor graphs, demonstrated on indoor multi-robot mapping scenarios.

For extreme environments, Chang et al. presented **LAMP 2.0** [11], a robust multi-robot SLAM system designed for operation in challenging large-scale underground environments. Developed for the DARPA Subterranean Challenge, LAMP 2.0 integrates a single-robot front-end adaptable to different odometry sources and LiDAR configurations with a scalable multi-robot front-end supporting both intra- and inter-robot loop closure detection. The system employs an outlier-resilient back-end equipped with robust optimization to handle perceptually-degraded conditions including darkness, dust, and self-similar scenes. Field validation across underground mines, tunnels, and caves spanning over 6 kilometers demonstrated LAMP 2.0's ability to maintain localization accuracy and map consistency for heterogeneous teams including aerial UAVs, ground UGVs, and legged quadrupeds under severe communication and sensing constraints.

Traditional multi-robot systems assume reliable communication infrastructure, yet real-world deployments frequently encounter intermittent connectivity, network partitions, and bandwidth variability. Miller et al. addressed these challenges through **MOCHA (Multi-robot Opportunistic Communication for Heterogeneous Collaboration)** [39], a framework enabling large-scale exploration without continuous communication requirements. MOCHA employs gossip-based protocols where robots opportunistically exchange information during brief communication windows, propagating map updates, artifact locations, and exploration frontiers throughout the team via epidemic-style dissemination.

The system coordinates heterogeneous aerial and ground robots with different communication ranges and mobility characteristics, where UAVs serve as mobile relays extending connectivity between ground robots. Field experiments with Boston Dynamics Spot robots and custom UAVs in forest canopies and urban environments demonstrated sustained performance despite 50% link loss rates, with information eventually reaching all team members through multi-hop gossip exchanges. MOCHA represents a paradigm shift from assuming continuous connectivity toward designing for intermittent communication as a fundamental system constraint.

While traditional SLAM passively maps explored regions, Active SLAM (A-SLAM) systems strategically plan robot trajectories to reduce map and pose uncertainty. Farhan et al. developed an **Efficient Multi-robot Active SLAM** [19] framework that extends single-robot A-SLAM to collaborative teams through intelligent frontier selection and priority-based coordination. The system incorporates a utility function leveraging frontier path entropy to evaluate candidate exploration goals, balancing information gain against travel cost and map uncertainty.

The framework employs a client-server architecture where lightweight nodes on each robot handle local execution while a central server manages global map fusion and goal assignment. To prevent starvation of low-priority robots, the system implements dynamic priority boosting when a robot's requests are repeatedly preempted. Validation through both Gazebo simulations and real TurtleBot experiments demonstrated 20-30% improvements in exploration time compared to standard frontier-based methods, with the approach achieving efficient coverage in both structured indoor environments and unstructured outdoor settings. The work highlights the benefits of coordinating exploration strategies across multi-robot teams rather than independent frontier assignment.

Recent advances in reinforcement learning have enabled adaptive task allocation strategies that handle complex operational constraints. Suzuki et al. addressed **multi-robot task allocation with infeasible tasks** [48], where some objects cannot be transported by any single robot due to weight or capability constraints. Their RL-based framework learns to detect infeasible assignments and dynamically reallocate

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

work to prevent deadlock, using Deep Q-Networks to optimize allocation policies across heterogeneous robot teams. By explicitly modeling task feasibility and incorporating it into the reward structure, the system avoids gridlock situations where robots indefinitely wait for impossible tasks, improving overall throughput in warehouse and logistics scenarios.

Extending beyond simple task-robot assignment, Ye et al. proposed a **two-stage reinforcement learning approach** [59] that decomposes resource allocation into activation and assignment phases. The first stage uses RL to decide which resources (robots, tools, workers) to activate based on predicted workload and activation costs, while the second stage assigns tasks to activated resources through optimization or secondary RL agents. This decomposition reduces the action space complexity while enabling explicit reasoning about resource costs, demonstrated through experiments in power grid management and logistics domains where activating resources incurs significant overhead.

Apart from individual protocol implementations, understanding protocol trade-offs is critical for deployment. Sanghvi et al. conducted an extensive comparative analysis of Wi-Fi, Bluetooth, Zigbee, and LoRa protocols for agricultural multi-robot coordination [44]. Simulations on the PARAM GANGA supercomputer evaluated SNR, Bit Error Rate (BER), data rate, range, and power consumption across various field sizes and robot densities. Results showed Wi-Fi excels in high-throughput short-range scenarios, Bluetooth suits moderate-range low-latency tasks, Zigbee balances power efficiency with mesh networking, and LoRa provides unmatched long-range communication at the cost of lower data rates. This work emphasizes that protocol selection must align with task requirements, environmental constraints, and energy budgets, particularly for battery-powered field robots operating over extended durations.

An often-overlooked but critical dimension of multi-robot systems is the interaction among multiple human operators supervising robot teams. Patel et al. investigated how direct and indirect communication modalities affect multi-human multi-robot interaction (MHMRI) [41]. Through a user study involving 18 operators controlling 9 robots in a transport task, they demonstrated that mixed communication (combining verbal interaction with interface-mediated transparency features) yielded the highest situational awareness, trust, and usability scores. The interface provided real-time visibility into robot states, peer actions, and task progress through on-robot panels, text logs, and object status widgets. This work emphasizes that as robot teams scale, interfaces must support both explicit human-to-human dialogue and implicit coordination through shared visual representations, reducing cognitive load and preventing out-of-the-loop performance problems.

Collaborative perception has become a cornerstone of modern multi-robot systems, enabling teams to share sensor data and fuse information for improved situational awareness. Nair explored collaborative perception frameworks for household cleaning and warehouse operations [40]. The study compared standalone perception (robots using only onboard sensors) against collaborative perception (robots augmented by overhead cameras and centralized fusion). Results showed that collaborative systems significantly reduced navigation time, improved obstacle detection under occlusion, and enabled better task allocation through global environmental awareness. The framework employed YOLOv8 for semantic segmentation and object detection, demonstrating how centralized perception processing can complement distributed robot control, particularly in structured environments like warehouses.

In logistics settings, Wang and Liu proposed a multimodal deep learning approach combining YOLOv5 for object detection, Proximal Policy Optimization (PPO) for action selection, and Asynchronous Advantage Actor-Critic (A3C) for multi-robot coordination [53]. Their system addressed path planning, task allocation, and dynamic scheduling in warehouse environments by integrating visual perception with reinforcement learning-based control. Experiments across multiple datasets showed improved task completion efficiency and adaptability to environmental changes, underscoring the value of tightly coupling perception and learning in collaborative robot teams.

Balancing computational loads between resource-constrained robots and powerful centralized servers remains a fundamental challenge in multi-robot systems. Bartolomei et al. introduced an agent-server architecture for multi-UAV coordination in partially unknown environments [5]. Their split design offloads computationally demanding tasks (global dense mapping, multi-robot path planning) to a central server while maintaining lightweight Visual-Inertial Odometry (VIO) and local obstacle avoidance onboard each UAV. The server fuses agent experiences through pose-graph optimization, constructs a global dense map using Voxblox, and performs hierarchical planning that assigns priorities to avoid inter-agent collisions. Experiments with four UAVs in photo-realistic industrial environments validated the approach's ability to handle partially overlapping waypoints and unknown map regions, addressing practical deployment constraints often overlooked in purely distributed methods.

Game-theoretic frameworks offer an alternative coordination paradigm by modeling multi-robot navigation as strategic interactions. Chandra et al. proposed decentralized social navigation using Game-Theoretic Control Barrier Functions (CBFs) [10]. Their approach combines CBF safety guarantees with Nash equilibrium solutions, enabling robots to navigate through constrained environments (narrow corridors, doorways) while respecting social norms and avoiding deadlocks. Unlike purely reactive methods, the game-theoretic formulation anticipates other agents' behaviors, leading to smoother trajectories and improved throughput in high-density scenarios.

Deep learning has increasingly been applied to multi-robot coordination challenges that resist traditional optimization approaches. Ye et al. introduced neural bipartite graph matching for active mapping [58], where a learned policy assigns robots to frontier regions by predicting the information gain from potential observations. Their Graph

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Neural Network architecture captures spatial relationships between robots and exploration frontiers, outperforming greedy and information-theoretic baselines in coverage efficiency and exploration time. This work demonstrates how learned coordination policies can implicitly balance exploration-exploitation trade-offs without explicit reward engineering.

Centralized Training with Decentralized Execution (CTDE) has emerged as an effective paradigm for multi-agent reinforcement learning. Yang et al. proposed learning macro-action-based policies via a centralized Q-network [56], enabling robots to coordinate on temporally extended actions rather than primitive controls. By hierarchically decomposing tasks into macro-actions (e.g., "patrol region A" rather than individual movement commands), their system reduces communication frequency and improves sample efficiency during training. Real-world experiments with mobile robots in warehouse environments showed 30% improvement in task completion time compared to flat action spaces.

Security and privacy concerns have motivated cryptographic approaches to multi-robot coordination. Kumar et al. developed communication-efficient path planning using Fully Homomorphic Encryption (FHE) [29], allowing robots to jointly plan paths without revealing their individual locations or goals to a central coordinator. While FHE introduces computational overhead, their optimizations (batch encryption, compressed trajectory representations) reduced encrypted communication to acceptable latencies for non-time-critical missions. This work opens pathways for multi-robot systems operating in adversarial or privacy-sensitive environments.

Range-based relative localization provides a minimal-sensing alternative to vision or GPS-based methods. Zhang et al. proposed distributed pose estimation using only inter-robot distance measurements [61], formulating the problem as distributed graph optimization. Their algorithm converges to accurate relative poses even with sparse connectivity and noisy ranging, demonstrated through experiments with ground robots in GPS-denied indoor environments. The approach is particularly valuable for underground, underwater, or radio-frequency-denied scenarios where traditional localization fails.

Collectively, these developments illustrate a paradigm shift in multi-robot collaboration from rigid, rule-based systems toward intelligent, interpretable, and communication-efficient frameworks that integrate perception, learning, and human cognition. This growing convergence of autonomy, learning, and human interaction is driving the next generation of scalable and trustworthy multi-robot systems.

This survey consolidates these advances, offering a taxonomy of collaboration strategies, communication models, and learning paradigms as shown in Fig. 1, followed by a synthesis of open research problems and directions for future work.

5. Communication and Coordination Frameworks

Coordination strategies range from centralized planning where a single brain assigns tasks to fully decentralized consensus-based approaches. Hybrid approaches (e.g., HMAS, COHERENT) combine global planners with local refinement, providing both scalability and fault tolerance [34].

Communication plays a central role: bandwidth, latency, and network topology determine how effectively robots can share situational awareness. Recent systems emphasize **sparsity**, sharing only high-value information such as loop closures in SLAM maps (Swarm-SLAM) [46] or feature vectors in GNN-based perception systems. Such selective communication improves scalability in bandwidth-limited scenarios.

Recent frameworks extend these concepts into real-world systems. Hu et al. [24] integrated distributed communication using the DDS protocol for stable multi-robot networking, while Yang et al. [57] employed ROS-based modules for coordinated mapping and task allocation. Similarly, Krizmancic et al. [27] demonstrated effective aerial-ground coordination through decentralized data exchange.

Beyond these conventional communication architectures, recent frameworks have begun expanding coordination beyond network protocols toward semantic and physical interaction. The **HMCF** [32] introduces a hierarchical communication model where each robot operates a local LLM-based controller for task verification, while a central assistant LLM manages dialogue, coordination, and human oversight. This represents a new form of *semantic communication*, where intent, reasoning, and task understanding are exchanged rather than only data packets. Conversely, the **Omnid Mocobots** [17] present a physical form of coordination known as *mechanical communication* where robots share information through forces transmitted via a shared payload instead of wireless links. These complementary paradigms broaden the definition of communication in multi-robot systems, linking cognitive-level reasoning with physical interaction to achieve more adaptive and human-aligned collaboration.

5.1. Concurrent Transmission for Low-Latency Coordination

Traditional multi-robot communication relies on asynchronous transmission, which suffers from arbitrary collisions, high latency, and significant energy waste. Bharadwaj et al. proposed a **concurrent transmission-based strategy** to achieve faster and more energy-efficient coordination [7]. The key innovation is a **split architecture** that physically separates the Networking Unit (NU) (responsible for concurrent transmission protocols) from the Computing and Control Unit (CCU) (handling robotics computations). This separation addresses the fundamental challenge that robotics computations introduce delays incompatible with the microsecond-level synchronization required for concurrent transmission.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

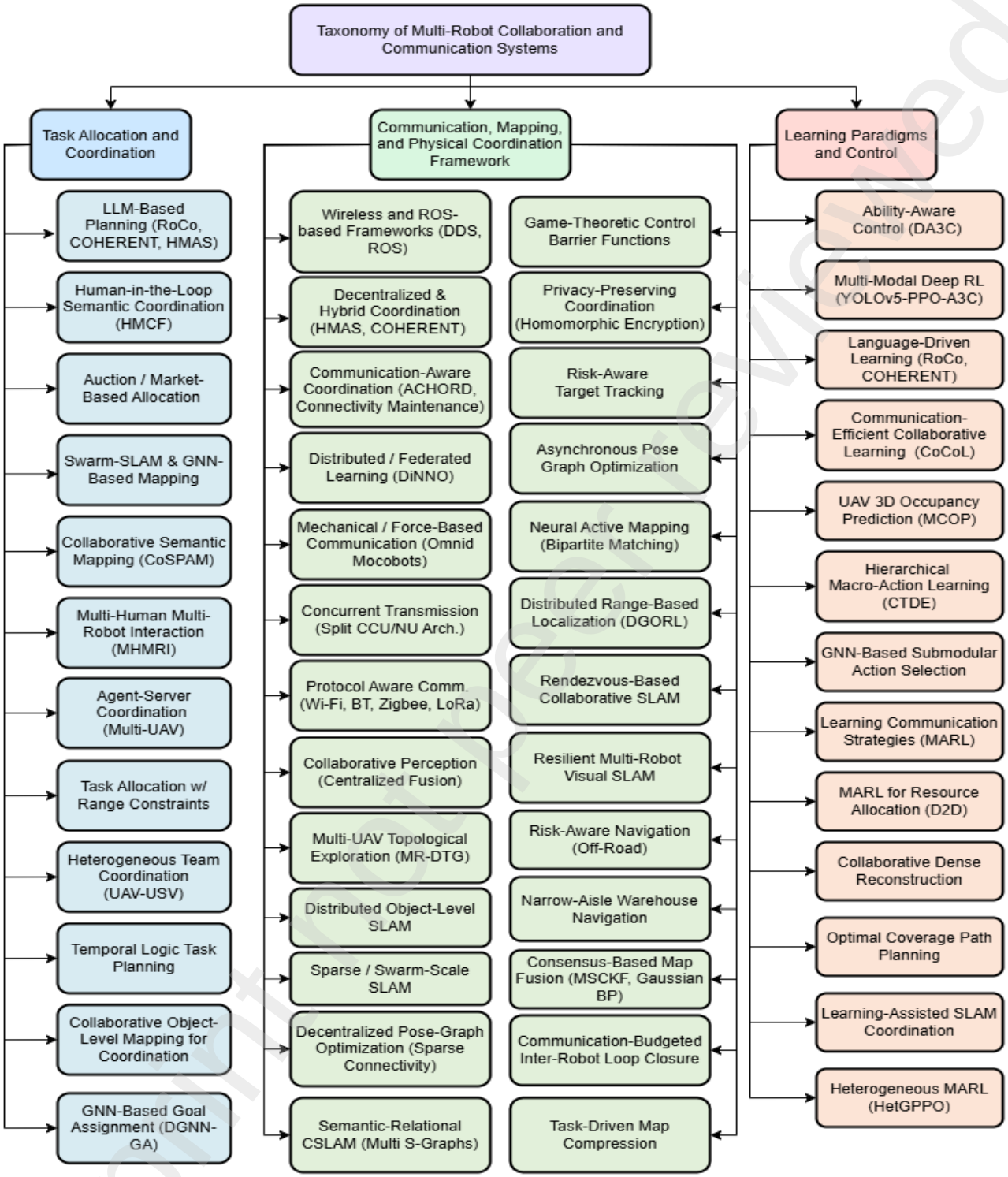


Figure 1: Taxonomy of Multi-Robot Collaboration and Communication Systems.

The NU runs the Glossy protocol, a well-known concurrent transmission algorithm that achieves network-wide data dissemination in milliseconds with high reliability. The CCU and NU coordinate through periodic serial communication, maintaining loose coupling that accommodates hardware heterogeneity. Real-world experiments with five two-wheel differential-drive mobile robots demonstrated centimeter-level spatial accuracy and millisecond-level temporal synchronization in a leader-follower formation control scenario over multi-hop connectivity. Radio duty-cycling remained very

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Table 1
Comparison of Wireless Communication Protocols for Agricultural Multi-Robot Coordination

Protocol	Data Rate	Range	Power	Best Use Case
Wi-Fi	Up to 6.9 Gbps	<50m (steep SNR drop beyond 70m)	High	Short-range, high-bandwidth tasks
Bluetooth	3 Mbps	∼30m	Low (5 mW idle)	Close-range robot-to-robot coordination
Zigbee	250 kbps	Up to 100m	Minimal	Medium-scale fields with mesh networking
LoRa	50 kbps	>1 km	Ultra-low (120 mW TX)	Sparse deployments over large areas

low, extending battery life significantly compared to asynchronous methods. This work establishes concurrent transmission as a viable and efficient communication paradigm for time-sensitive multi-robot applications.

5.2. Protocol Selection and Performance Trade-Offs

While concurrent transmission addresses latency and energy in controlled settings, real-world deployments (especially in agriculture and outdoor environments) require careful protocol selection tailored to operational constraints. Sanghvi et al. conducted a comprehensive performance analysis of Wi-Fi, Bluetooth, Zigbee, and LoRa for agricultural multi-robot coordination [44]. Using the PARAM GANGA supercomputer, they simulated a 100×100m agricultural field with four autonomous robots, incorporating Gaussian-distributed sensor noise and Signal-to-Noise Ratio (SNR) degradation models based on distance and interference. Table 2 summarizes the key findings.

The study demonstrated that protocol choice must account for field size, robot density, task frequency, and energy budgets. For pesticide spraying over large fields with infrequent updates, LoRa's long range and low power dominate. For real-time obstacle avoidance in orchards, Bluetooth or Zigbee provide sufficient throughput and lower latency. This comparative framework provides practitioners with empirical guidance for protocol deployment in agricultural and similar outdoor multi-robot scenarios.

5.3. Simulation Middleware for Heterogeneous Multi-Robot Communication

Dey et al. introduced **SynchroSim**, an integrated co-simulation middleware that bridges physics-based and network-based simulators for heterogeneous multi-robot environments [14]. The framework couples Gazebo (for motion and environment dynamics) with NS-3 (for network communication), enabling synchronized testing of UAV-Unmanned Ground Vehicle (UGV) communication in both Line-of-Sight (LOS) and Non-Line-of-Sight (NLOS) conditions. A key feature of SynchroSim is the *velocity-difference-driven adjustable window* method, which dynamically adjusts the synchronization interval based on the relative speeds of communicating robots. This approach reduced packet loss and delay by approximately 10-11% compared to fixed-window synchronization.

The study validates both simulation and real-world scenarios using Duckiebots, demonstrating low delay and negligible packet loss in master-based Robot Operating System (ROS) communication setups. Although the system performed well in small-scale tests, scalability and adaptability to 5G and outdoor conditions remain open challenges. Future extensions proposed by the authors include migrating to ROS2 with Data Distribution Service (DDS) for decentralized communication, integrating AI-based predictive synchronization, and expanding to outdoor heterogeneous clusters [14].

5.4. Communication-Aware Coordination Under Intermittent Connectivity

Recent work has explored communication-aware frameworks that allow multi-robot teams to adaptively maintain connectivity in dynamic or degraded network environments. Saboia da Silva et al. proposed **ACHORD** (Communication-Aware Multi-Robot Coordination with Intermittent Connectivity), an architecture designed for challenging environments such as underground tunnels and mines [45]. ACHORD integrates low-level networking metrics such as signal-to-noise ratio (SNR), bandwidth, and packet delay with high-level decision-making to guide exploration and communication actions.

When signal strength weakens, robots autonomously deploy small relay radios to extend communication range, forming a dynamic mesh network that keeps the team connected. The system classifies outgoing data into three categories that are *key*, *mission-critical*, and *time-sensitive*, and prioritizes transmission accordingly. For instance, safety messages and collision alerts are sent immediately, while regular map updates are transmitted when bandwidth permits.

Tested in the **DARPA Subterranean Challenge** and real limestone mine environments using **Boston Dynamics Spot** and **Clearpath Husky** robots, ACHORD demonstrated stable data transfer beyond 150 meters with sustained throughput over 20 Mbps. Compared to its predecessor CHORD, ACHORD reduced data congestion and latency while enabling autonomous radio deployment and link recovery after disconnection. Although scalability and energy efficiency remain open challenges, ACHORD represents a major step toward robust multi-robot collaboration in communication-denied environments.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

5.5. Task-Driven Map Compression for Communication-Constrained Navigation

Complementing relay-based approaches like ACHORD, Psomiadis et al. introduced a **communication-aware iterative map compression** framework that addresses bandwidth constraints at the information level rather than the network level [42]. In their Sensor-Actor architecture, a mobile sensor robot explores an environment and must transmit its local map to a remote actor robot for path planning. Rather than sending the full occupancy grid, the sensor selects from a library of *abstraction templates* (e.g., quadtree decomposition, random sampling, uniform downsampling) and compresses the map into a compact representation that preserves task-relevant structure.

The key innovation is an *iterative Kalman filter decoder* at the actor side that reconstructs the full map from the compressed representation. By modeling the compression-reconstruction process as a state estimation problem, the framework naturally handles uncertainty: the actor maintains a belief over the true map and can request additional refinement from the sensor when reconstruction confidence is low. This iterative protocol allows the system to adaptively allocate communication bandwidth based on map complexity and task requirements.

Evaluated on both Mars terrain datasets (HiRISE orbital imagery) and Earth environments, the framework achieves up to 98% reduction in communicated data while maintaining near-optimal path quality (within 5% of paths computed on full-resolution maps). The abstraction template selection mechanism enables task-driven compression: for long-range navigation, coarse templates suffice, while fine-grained templates are reserved for regions near obstacles or narrow passages. This approach demonstrates that communication efficiency in multi-robot navigation can be achieved through intelligent information compression rather than solely through network-level optimization.

5.6. Task Allocation Under Communication and Energy Constraints

Resource constraints fundamentally shape multi-robot coordination strategies. Tang et al. addressed task allocation and path planning with maximum communication range constraints [49], formulating the problem as mixed-integer linear programming with explicit connectivity requirements. Their approach ensures that robots remain within communication range of at least one neighbor throughout mission execution, preventing network fragmentation. Experiments with mobile robot teams validated the method's ability to balance task optimality with connectivity maintenance, though scalability beyond 10-15 robots remains challenging due to computational complexity.

Learning-based approaches offer an alternative to optimization-based task allocation under communication constraints. Goarin and Loianno proposed **DGNN-GA** (Decentralized Graph Neural Network for Goal Assignment), a GNN-based framework that solves the multi-robot goal assignment problem in a fully decentralized manner [22]. Unlike auction-based methods such as CBAA that require multiple communication rounds to converge, DGNN-GA formulates goal assignment as edge classification on a heterogeneous graph containing both robot and goal nodes. Each robot runs a local GNN that processes only neighborhood information, requiring a fixed number of message-passing rounds independent of team size. The framework outperforms both the Consensus-Based Auction Algorithm (CBAA) and the Decentralized Hungarian-Based Algorithm (DHBA) under severely restricted communication (limited message rounds), while demonstrating zero-shot generalizability from training on 20-robot teams to deployment on 50-robot teams. Real-world validation on quadrotor platforms confirmed the sim-to-real transfer, with assignment quality within 5% of the centralized Hungarian optimal solution.

Heterogeneous teams introduce additional coordination challenges stemming from varied capabilities and sensing modalities. Kulathunga and Klimchik developed a system integrating UAVs and unmanned surface vehicles (USVs) for water quality monitoring [28]. UAVs provide aerial reconnaissance to identify sampling locations, while USVs collect water samples and perform in-situ measurements. Their coordination strategy dynamically reallocates tasks based on battery levels, weather conditions, and mission priorities, demonstrating 40% improvement in coverage efficiency compared to homogeneous teams in field trials across lakes and rivers.

5.7. Adversarial and Safety-Critical Coordination

Multi-robot systems deployed in adversarial settings must balance task completion with threat avoidance. Sung et al. introduced target tracking with sensing and communication danger zones [47], where certain regions disable sensors or communication links. Their approach formulates tracking as a partially observable Markov decision process (POMDP) with explicit risk-aware planning, enabling robots to maintain target observation while avoiding or minimizing exposure to danger zones. Simulations with ground robots tracking moving targets showed significant improvements in tracking duration and safety compared to risk-agnostic methods.

5.8. Distributed SLAM Communication Strategies

Efficient map sharing lies at the heart of collaborative SLAM, where naive approaches that exchange full pose graphs quickly exhaust available bandwidth. **Swarm-SLAM** [30] addresses this challenge through spectral prioritization of inter-robot loop closures. Rather than sharing all detected place recognition candidates, Swarm-SLAM analyzes the pose graph Laplacian matrix to identify loop closures that maximally increase graph connectivity and reduce map uncertainty.

The spectral method computes eigenvalues of the Laplacian for candidate loop closures, selecting those that produce the largest increase in algebraic connectivity (the second-smallest eigenvalue, also known as the Fiedler value). This criterion ensures that shared loop closures provide maximum information gain for the team's collective map, avoiding

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

redundant constraints that offer minimal uncertainty reduction. By transmitting only spectrally-prioritized loop closures, Swarm-SLAM achieves up to 60% reduction in communication overhead while maintaining near-centralized mapping accuracy. Real-world validation in the DARPA Subterranean Challenge demonstrated scalability to large teams exploring GPS-denied underground environments with intermittent connectivity.

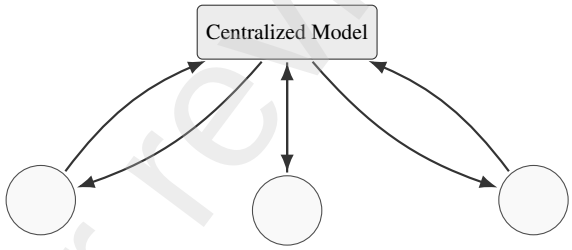
Complementing geometric SLAM approaches, **SlideSLAM** [35] introduces metric-semantic maps for heterogeneous multi-robot teams. The key insight is that object-level representations provide viewpoint-invariant place recognition, critical for heterogeneous teams with different sensor configurations and perspectives. SlideSLAM's front-end performs instance segmentation using either RGB-D cameras or LiDARs, extracting object landmarks rather than geometric features. The back-end optimizes robot trajectories and object positions through custom factor graph optimization, with semantics-driven place recognition algorithms matching object configurations across robots.

Object-level representations have also been explored through distributed variational inference [9], where robots maintain probabilistic beliefs over object poses and exchange information via message-passing to achieve globally consistent maps. This approach provides uncertainty quantification critical for decision-making under ambiguity.

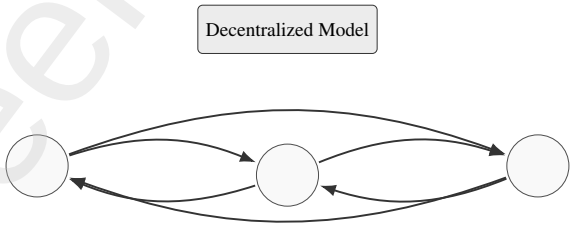
For inter-robot loop closure, SlideSLAM develops two complementary algorithms: SlideMatch uses exhaustive search over transformation space with zero-centered map preprocessing to reduce search regions, while SlideGraph performs descriptor-based matching using triangle descriptors derived from object positions and classes. Both algorithms leverage the informativeness of semantic representations: matching "desk, chair, monitor" configurations is more robust to viewpoint changes than matching geometric features. Validated across aerial, ground, and legged platforms in diverse indoor and outdoor environments, SlideSLAM demonstrates that semantic abstraction enables effective collaboration among heterogeneous teams where traditional geometric methods fail due to sensor and viewpoint differences.

Advancing beyond object-level representations, **Multi S-Graphs** [21] introduces a hierarchical semantic-relational approach to decentralized collaborative SLAM. Rather than exchanging raw point clouds or even object-level landmarks, Multi S-Graphs constructs *situational graphs* that encode the environment as a hierarchy of structural elements: walls, rooms, and floors. Each robot maintains a local S-Graph and generates compact *room-based descriptors* for inter-robot loop closure detection. When two robots' room descriptors match, they exchange only the relevant subgraph rather than the full map, achieving over 95% reduction in data exchange compared to point-cloud-based methods.

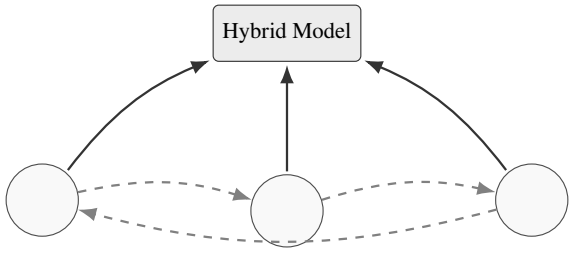
The key insight is that room-level semantic abstractions provide highly discriminative yet compact representations for place recognition: matching "office with 4 walls and 2 doors" is both more robust and more bandwidth-efficient than matching geometric point clouds. A graph distillation mechanism further compresses the situational graph by merging redundant structural elements, reducing communication overhead while preserving topological accuracy. Experiments in multi-floor building environments demonstrated over 90% improvement in localization accuracy compared to state-of-the-art decentralized SLAM baselines, with the hierarchical representation enabling efficient loop closure across robots exploring different floors of the same building. Multi S-Graphs demonstrates that structured semantic-relational representations can simultaneously improve both communication efficiency and mapping accuracy in collaborative SLAM.



(a) Centralized: all robots communicate through a single coordinator.



(b) Decentralized: robots exchange information directly with peers.



(c) Hybrid: combines global planning with local peer-to-peer collaboration.

Figure 2: Communication and Coordination Models in Multi-Robot Systems.

6. Representative Frameworks and Applications

This section reviews the state-of-the-art literature on multi-robot collaboration and communication, emphasizing recent advances that combine learning-based strategies, decentralized mapping, and natural-language-driven coordination. The reviewed works span task allocation, communication architectures, representative applications, and learning paradigms.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

6.1. Task Allocation and Language-Driven Coordination

Task allocation lies at the core of collaborative robotics. Traditional methods rely on market-based or auction-based allocation strategies, where tasks are distributed to robots based on bids and costs. Recent research explores natural-language-driven coordination, allowing robots to negotiate and decompose tasks in ways that are interpretable to human supervisors.

Zhao et al. introduced **RoCo**, a dialectic multi-robot collaboration framework where robots controlled by Large Language Models (LLMs) communicate in natural language to plan tasks collaboratively [37]. RoCoBench, a benchmark comprising six collaborative tasks such as rope manipulation and object sorting, demonstrated that conversational task planning improves team adaptability and interpretability. An important contribution of RoCo is its feedback loop: if potential collisions or infeasible plans are detected, robots iteratively refine their subtask plans until convergence. This capability enables seamless human-in-the-loop participation.

Chen et al. investigated **HMAS** (Hybrid Multi-Agent Systems), comparing fully decentralized, fully centralized, and two hybrid frameworks using LLMs for task planning [13]. Their results showed that HMAS where a central planner proposes a plan and local agents validate and provide feedback, achieves higher success rates and scalability than purely centralized or decentralized systems. The framework reduces excessive dialogue and token usage common in fully decentralized methods, making it more practical for large teams.

Liu et al. presented **COHERENT**, a hierarchical coordination system that assigns tasks to heterogeneous robot teams (drones, mobile manipulators, quadrupeds) using a Proposal-Execution-Feedback-Adjustment (PEFA) cycle [34]. COHERENT demonstrated high success rates (97.5%) across 100 long-horizon tasks, outperforming baseline planners in both simulation and real-world experiments. Its strength lies in dynamically adjusting plans based on robot feedback, which is essential for mixed-team settings.

Together, these works indicate a growing trend toward LLM-powered task planning, offering interpretability and flexibility, though challenges remain in computational cost, latency, and robustness against sensor noise.

Building on these language-driven systems, the **HMCF (Human-in-the-loop Multi-Robot Collaboration Framework)** [32] extends LLM-based planning by introducing explicit human oversight into the coordination loop. In HMCF, each robot runs a local LLM agent responsible for verifying task feasibility, while a central assistant LLM manages global planning, communication, and human dialogue. This hierarchical structure improves interpretability, reduces coordination errors, and maintains adaptability across heterogeneous robot teams. Real-world experiments demonstrated HMCF's ability to handle new tasks without retraining and to outperform earlier LLM-based planners such as RoCo and COHERENT in terms of success rate and task completion time.

Complementing semantic collaboration, the **Omnid Mocobots** [17] framework explores physical multi-robot cooperation through *mechanical communication*. Rather than exchanging wireless messages, Omnid Mocobots use forces transmitted via a shared payload as the medium of coordination. Each mocobot combines an omnidirectional mobile base with a compliant Delta manipulator, allowing intuitive and safe co-manipulation of large or articulated objects. This design enables both robots and humans to sense and respond to each other's intentions through passive compliance, extending the concept of communication from digital data to physical interaction.

6.2. Communication Architectures and Decentralized Mapping

Efficient communication is critical for multi-robot systems, especially under bandwidth limitations. Graph Neural Network (GNN)-based perception pipelines allow robots to share compact feature embeddings rather than raw sensor data, resulting in scalable collaborative perception [63]. By representing each robot as a node in a graph and exchanging messages with neighbors, GNNs fuse multi-view information to enhance detection performance even under occlusion or partial observability.

A major development in distributed SLAM is **Swarm-SLAM**, a fully decentralized and resource-efficient collaborative mapping framework. It introduces spectral prioritization for inter-robot loop closures, ensuring that only the most informative constraints are shared while maintaining high localization accuracy. Real-world experiments demonstrated that Swarm-SLAM achieves near-centralized map quality with up to 60% fewer transmitted loop closures, making it practical for GPS-denied, large-scale environments.

Gusrialdi et al. introduced a connectivity-aware coordination framework where robots simultaneously perform their tasks while preserving strong communication links within the team [23]. Rather than assuming that shorter distances guarantee reliable communication, their system learns real wireless signal strength using Gaussian Process prediction and enforces connectivity using Control Barrier Functions. Robots follow their nominal task trajectories and receive minimal corrective motion only when communication is at risk. Simulations with up to 20 robots showed improved connectivity robustness and fewer task interruptions compared to distance-only baselines.

These approaches collectively illustrate a shift toward **communication-efficient collaboration**, where only essential messages such as loop closures, feature embeddings, or network-recovery actions are exchanged. This paradigm enhances scalability by reducing bandwidth consumption while preserving shared situational awareness.

Complementary system-level studies have applied these principles in practice. Hu et al. [24] employed distributed DDS networking for real-time multi-robot mapping, while Yang et al. [57] demonstrated coordinated exploration using ROS-based communication modules.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

6.3. Applications of Multi-Robot Collaboration

6.3.1. Search and Rescue (SAR)

Multi-robot SAR systems combine heterogeneous platforms e.g. UAVs, ground vehicles, and aquatic robots to perform fast, reliable victim localization and mapping in challenging environments [43]. Active perception techniques enable robots to move strategically to improve viewpoints, while connectivity-aware path planning ensures communication is maintained. Chatziparaschis et al. demonstrated UAV-UGV collaboration where a UAV performs aerial SLAM and shares a 2.5D map with a ground humanoid robot for navigation and human detection [12].

6.3.2. Security and Counter-UAV

In security applications, collaborative systems have been designed to detect, track, and neutralize aerial intruders. The **MARCUS** system integrates a ground robot with LiDAR, a pursuer UAV with stereo vision, and an interceptor UAV equipped with a net launcher [4]. The system performs fully autonomous detection, verification, and interception, with a human operator providing final authorization. Field trials confirmed its ability to capture real intruder drones safely, demonstrating the potential of multi-robot collaboration in security-critical domains.

6.3.3. Exploration and Navigation

Yu et al. proposed **Co-NavGPT**, a cooperative exploration system where a Vision-Language Model (VLM) guides multiple robots to efficiently explore unknown environments [1]. The VLM reasons about semantic cues (e.g., "chairs are likely in living rooms") and assigns robots to unexplored frontiers, resulting in faster target discovery compared to heuristic planners. This method demonstrates the growing role of semantic reasoning in coordinated exploration.

Building on the importance of semantic understanding, **CoSPAM** introduces a collaborative semantic mapping and simultaneous path planning framework for efficient exploration of large-scale environments [36]. Each robot creates a local map using RGB-D perception and sampling-based motion planning, while a central server fuses keyframes into a global semantic OctoMap to ensure consistent environmental understanding. By assigning non-overlapping navigation goals, the framework minimizes redundant motion and improves team efficiency. Experiments demonstrated up to 56% faster coverage and shorter travel distances compared to classical frontier-based exploration, highlighting the benefits of integrated mapping and collaborative navigation.

6.3.4. Wireless Multi-Robot Collaboration

Hu et al. developed a **Wireless Multi-Robot Collaboration System** that integrates communication, perception, control, and planning within a distributed wireless network [24]. Their framework allows UAVs, wheeled robots, and quadrupeds to share mapping and navigation data using the DDS protocol. Each robot performs local SLAM and transmits selected keyframes to a central node, which fuses them into a global 3D OctoMap while filtering dynamic objects. The system employs an RRT-based global planner to avoid low-signal regions and dynamically adjust communication routes. Kubernetes-based containerization ensures scalability across agents. Experiments in simulation and semi-physical environments demonstrated mapping errors under 2 cm and a 73% reduction in task completion time compared to single-robot setups. Limitations include reliance on centralized fusion, lack of large-scale outdoor tests, and a limited range of tasks, leaving room for distributed enhancements.

6.3.5. ROS-Based Multi-Robot Control for Disaster Environments

Yang et al. introduced a **ROS-based Multi-Robot Control System** to support autonomous disaster exploration and mapping by multiple ground robots [57]. The system emphasizes ease of use for non-expert operators through a graphical RQT interface. Users can assign waypoints on a 2D occupancy map, and each robot automatically plans and executes its trajectory using A* search. The Consensus-Based Bundle Algorithm (CBBA) assigns tasks among robots based on spatial proximity and workload. Local maps generated via LIDAR-based SLAM are merged into a global view through ROS mapping utilities. Real-world experiments with Turtlebot robots validated the system's effectiveness for collaborative exploration, achieving accurate map merging and minimal operator input. However, scalability remains limited due to communication constraints and the homogeneous nature of the robot team.

6.3.6. Cooperative Aerial-Ground Multi-Robot Construction

Krizmancic et al. presented a **Cooperative Aerial-Ground Multi-Robot System** designed for autonomous construction tasks [27]. The system coordinates UAVs and a UGV to detect, transport, and assemble colored bricks during a wall-building task. The coordination follows a decentralized structure using the TAEMS task model and Generalized Partial Global Planning (GPGP), which dynamically assigns roles and synchronizes progress. The UGV provides brick localization and placement feedback, while UAVs handle manipulation and positioning. Simulation results under the MBZIRC setup showed that this framework achieves near-optimal task allocation, within 12% of an MILP-based optimal scheduler, while maintaining high fault tolerance and scalability. Although physical construction tests remain limited, this study highlights the potential for using heterogeneous aerial-ground teams in automated construction and assembly missions.

6.3.7. Multi-Domain Ocean Monitoring with UAV-USV Collaboration

Manley et al. proposed a hybrid multi-domain robotic framework combining Uncrewed Aerial Vehicle (UAV) and Uncrewed Surface Vehicles (USV) for ocean observation and marine domain awareness [38]. The system integrates

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Seahawk Robotics' F4 amphibious drone with Open Ocean Robotics' DataCruiser™ USV, enabling extended-range, sustainable, and persistent marine sensing operations.

The UAV provides aerial visibility, while the USV serves as a mobile base for recharging, data offloading, and communication. A 30 m winch mechanism allows the UAV to deploy passive acoustic sensors for underwater monitoring. The DataCruiser's solar-powered and battery-supported energy system enables missions lasting weeks to months, supporting AI-driven detection, classification, and event response. The UAV can autonomously take off, land, and share sensor data with the USV using onboard AI for visual and acoustic fusion.

This collaboration introduces a scalable ocean monitoring network that supports long-duration missions, reduced operational cost, and minimal environmental impact. Limitations remain in autonomous landing during rough sea states and real-time data latency, with full system integration and field trials expected in 2025-2026 [38].

6.3.8. Human-Robot Co-Manipulation in Warehouses

The **Omnid Mocobots** [17] represent a new paradigm for human-robot collaboration in shared workspaces such as warehouses and manufacturing facilities. Each mocobot integrates an omnidirectional mobile base with a compliant Delta manipulator, enabling safe and intuitive physical interaction with humans during cooperative lifting and object transport tasks. Unlike traditional systems that rely solely on wireless communication, the mocobots use *mechanical communication*, transmitting intent through the forces applied to a shared payload. This allows both humans and robots to sense each other's actions in real time, improving coordination and reducing cognitive load. Experimental results demonstrated stable force sharing, minimal delay in motion adaptation, and robust co-manipulation of large objects, highlighting the potential of physically coupled multi-robot systems in warehouse logistics and material handling.

6.3.9. Multi-Human Multi-Robot Interaction and Interface Design

As multi-robot systems scale, the role of multiple human operators becomes increasingly important. Patel et al. explored how communication modalities among human operators affect system performance in multi-human multi-robot interaction (MHMRI) scenarios [41]. Their user study involved 18 participants (organized in teams of 2) supervising 9 mobile robots in a gamified object transport task over 8-minute sessions.

The study compared four communication conditions:

1. **No Communication (NC)**: Operators could not communicate or see each other's actions.
2. **Direct Communication (DC)**: Verbal and gestural interaction allowed, but no interface-mediated information sharing.
3. **Indirect Communication (IC)**: Interface provided transparency features (robot status panels, action logs, object panels) but no direct operator dialogue.
4. **Mixed Communication (MC)**: Both direct and indirect modalities available.

Interface transparency features included:

- **On-robot status panels** displaying current state and assigned tasks
- **Robot panel** showing functional/faulty robots with color-coded task engagement
- **Text-based action log** broadcasting operator manipulations of virtual objects
- **Object panel** indicating which objects were being moved or locked by which operator

Results demonstrated that **mixed communication (MC) significantly outperformed other conditions** across multiple metrics:

- **Situational Awareness (SART)**: MC scored highest, ensuring operators understood teammate intentions, robot states, and task progress.
- **NASA-TLX Workload**: MC distributed cognitive load more evenly between operators.
- **Trust**: MC fostered greater confidence in the system and co-operator reliability.
- **Usability**: Operators preferred MC, citing better coordination and reduced conflicts.

The study revealed that indirect communication alone (IC) improved awareness but lacked the flexibility of direct dialogue for rapid replanning. Direct communication alone (DC) enabled quick exchanges but suffered from incomplete situational awareness when operators focused on different areas. Mixed communication combined the strengths of both, supporting proactive information sharing (direct) and passive awareness through interface transparency (indirect). This work underscores that as robot teams grow, operator interfaces must facilitate both explicit human-to-human communication and implicit coordination through shared visual representations to prevent out-of-the-loop performance problems.

6.3.10. Collaborative Perception in Household and Warehouse Environments

Nair presented a comprehensive study on collaborative perception (CP) frameworks for multi-robot household cleaning and warehouse operations [40]. The work contrasts **standalone perception (SP)** (where each robot relies solely on onboard sensors) with **collaborative perception (CP)** (where robots are augmented by environment-embedded sensors transmitting data to a centralized server for fusion and decision-making).

Case Study 1: Household Cleaning Robots

The household scenario involved three robot types: a Robotic Vacuum Cleaner (RVC), a Staircase Cleaning Robot (SCR), and a Trash Picking Robot (TPR). Three overhead

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

cameras monitored different sections of the living space, streaming RGB data to a server running YOLOv8 models for:

- **Object detection** to identify dynamic obstacles (humans, pets)
- **Semantic segmentation** to delineate cleanable floor areas

The server issued commands based on detected conditions:

- If humans/pets detected → pause cleaning operations
- If trash detected → dispatch TPR first, then resume RVC/SCR operations
- Segment and assign cleanable areas dynamically to avoid overlaps

This collaborative approach enabled:

- **Safety**: Robots halted when humans entered the scene, preventing collisions.
- **Efficiency**: Task prioritization (trash removal before vacuuming) reduced redundant operations.
- **Coverage**: Overhead cameras provided a global view, eliminating blind spots inherent in robot-mounted sensors.

Case Study 2: Autonomous Mobile Robots (AMRs) in Warehouses

The warehouse scenario simulated multiple racks with AMRs carrying pallets autonomously. Two configurations were tested:

- **Standalone Perception (SP)**: Robots used only onboard LiDAR for obstacle detection and path planning.
- **Collaborative Perception (CP)**: Overhead LiDARs in each aisle transmitted real-time occupancy grids to a server, which fused data and provided AMRs with enhanced situational awareness beyond their sensor range.

Comparative Results:

- **Task Completion Time**: CP reduced average completion time by approximately 25% due to early obstacle detection and proactive rerouting.
- **Path Efficiency**: CP-enabled AMRs avoided last-minute braking and detours, resulting in smoother trajectories and lower energy consumption.
- **Collision Avoidance**: CP eliminated near-miss events caused by occlusions (e.g., robots hidden behind rack corners), which were frequent in SP mode.

The study demonstrated that collaborative perception is particularly advantageous in **structured environments with known layouts**, where strategic sensor placement can compensate for individual robot limitations. However, the reliance on centralized processing introduces potential bottlenecks and single points of failure, suggesting future work should explore distributed fusion or edge computing architectures for large-scale deployments.

6.3.11. Logistics and Warehousing with Deep Reinforcement Learning

Wang and Liu addressed multi-robot collaborative operation in logistics warehousing through a novel integration of computer vision and reinforcement learning [53]. Their **YOLOv5-PPO model optimized with A3C** combines:

- **YOLOv5**: Real-time object detection for identifying goods, obstacles, and navigation targets
- **Proximal Policy Optimization (PPO)**: Action policy learning for path planning and dynamic task execution
- **Asynchronous Advantage Actor-Critic (A3C)**: Coordinating multiple robots in parallel training environments

System Architecture:

1. **Perception Module**: YOLOv5 processes camera feeds to detect and classify objects in the warehouse.
2. **Decision Module**: PPO trains each robot's policy to navigate, avoid obstacles, and execute tasks based on perceived environment state and assigned goals.
3. **Coordination Module**: A3C enables parallel training across multiple robots, with each agent contributing experiences to update a shared policy network, accelerating convergence and improving robustness.

Key Innovations:

- **Multi-Modal Fusion**: Integrates visual perception (YOLOv5), time-series sensor data, and robot position information into a unified state representation.
- **Adaptive Scheduling**: Robots dynamically adjust task priorities and paths based on environmental changes (e.g., unexpected obstacles, demand fluctuations).
- **Scalable Training**: A3C's asynchronous updates allow efficient training even with heterogeneous robot hardware, as each agent can train at different speeds without blocking others.

Experimental Results: Evaluated on four datasets (AI4 Logistics, Multi-Robot Warehouse, ROS-based Logistics Simulation, RoboCup Logistics League), the model achieved:

- **Multi-Modal Fusion Score**: 96.18 (AI4 Logistics), significantly outperforming baselines (Orr & Dutta: 89.83, Chen et al.: 89.54)

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Table 2
Communication Protocol Performance Comparison for Agricultural Multi-Robot Systems

Protocol	Range	Data Rate	TX Power	RX Power	Idle Power
Wi-Fi	<70m	6.9 Gbps	400mW	200mW	50mW
Bluetooth	∼30m	3 Mbps	50mW	25mW	5mW
Zigbee	∼100m	250 kbps	30mW	20mW	3mW
LoRa	>1km	50 kbps	120mW	40mW	1mW

- **Consistency**: High inter-modal agreement, ensuring reliable decision-making across visual and sensor inputs
- **Adaptability**: Superior performance in transfer learning metrics, indicating generalization to novel warehouse configurations
- **Task Efficiency**: Reduced task completion time and improved resource utilization compared to standalone Deep Q-Network (DQN), Convolutional Neural Network (CNN), and Trust Region Policy Optimization (TRPO) baselines

The work demonstrates that combining state-of-the-art object detection with multi-agent reinforcement learning creates a powerful framework for logistics automation, addressing challenges in perception, coordination, and real-time decision-making. Future directions include extending the system to handle more complex manipulation tasks and integrating human-in-the-loop supervision for safety-critical operations.

6.3.12. Agricultural Multi-Robot Systems with Protocol-Aware Coordination

Sanghvi et al. developed a hierarchical multi-robot framework for precision agriculture, specifically targeting autonomous pesticide spraying across large fields [44]. The system architecture consists of three layers:

1. **Robot Layer**: Autonomous ground robots equipped with GPS, pesticide residue sensors, monitoring sensors, and cameras.
2. **Communication Gateway Layer**: Aggregates sensor data from robots using selected Internet of Things (IoT) protocols (Wi-Fi, Bluetooth, Zigbee, or LoRa) and relays formatted messages to a centralized server.
3. **Centralized Server Layer** (simulated on PARAM GANGA supercomputer): Processes sensor data, optimizes robot paths using graph-based algorithms, allocates tasks dynamically, and transmits action commands back to robots.

Protocol Performance Analysis:
The study systematically compared four protocols across key metrics (see Table 2).
Simulation Methodology:

- **Field Size**: 100×100m grid
- **Robot Configuration**: 4 autonomous robots, initial positions at grid corners, dynamic inter-robot distances (20m baseline)
- **Noise Modeling**: Gaussian-distributed sensor noise, SNR degradation proportional to distance and interference
- **Performance Metrics**: SNR vs. distance, BER vs. SNR, power consumption trends, task completion time
- **Simulation Iterations**: 30 independent trials per protocol, 1,000 time steps or full task completion

Key Insights:

- **Field Size Dictates Protocol**: For small orchards (<50m), Wi-Fi or Bluetooth suffice. For medium fields (50-150m), Zigbee's mesh networking provides robust coverage. For expansive crop fields (>200m), only LoRa maintains connectivity without relay nodes.
- **Task Frequency Matters**: High-frequency updates (obstacle avoidance, real-time path adjustments) favor low-latency protocols (Bluetooth, Zigbee). Infrequent updates (periodic status reports, post-task summaries) tolerate LoRa's higher latency.
- **Energy Constraints**: Battery-powered robots benefit from LoRa's ultra-low idle power for long-duration missions (days to weeks). Rechargeable robots with frequent charging access can afford Wi-Fi's higher consumption for better performance.

This work provides a practical decision framework for agricultural robotics practitioners, demonstrating that there is no one-size-fits-all protocol. Instead, system designers must match communication infrastructure to specific operational requirements, leveraging high-performance computing resources to validate protocol performance before field deployment.

6.3.13. Multi-UAV Exploration with Dynamic Topological Graphs

Dong et al. proposed MR-DTG (Multi-Robot Dynamic Topological Graph), a decentralized framework for fast and communication-efficient UAV exploration [15]. The key innovation is maintaining a lightweight topological graph representation of the environment shared among UAVs through

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

spectral sparsification, dramatically reducing communication overhead while preserving exploration efficiency.

Technical Approach:

- Each UAV constructs a local Dijkstra tree covering explored regions within its sensor range
- Voronoi partition assigns exploration regions to UAVs based on graph distance rather than Euclidean distance, accounting for obstacles and terrain constraints
- Selective sharing of topological graph updates (only significant changes) reduces bandwidth by 60% compared to full map exchange
- Hierarchical planning: global exploration via Voronoi frontiers, local obstacle avoidance via model predictive control

Real-world experiments with 6 UAVs in complex indoor and outdoor environments demonstrated:

- 40% faster exploration compared to frontier-based baselines
- Near-linear scalability up to 15 UAVs in simulation
- Robust performance under 20% packet loss and 500ms communication delays

The MR-DTG framework showcases how graph-theoretic representations can enable efficient multi-robot exploration without requiring dense inter-robot communication or centralized coordination.

6.3.14. Multi-UAV Collaborative Occupancy Prediction

Lin et al. introduced MCOP (Multi-UAV Collaborative Occupancy Prediction), addressing the unique challenges of UAV perception at altitudes 10× higher than ground vehicles [33]. Unlike bounding-box detection, occupancy prediction provides complete 3D scene representation including free space, occupied regions, and semantic labels.

Key Technical Contributions:

- **Spatial-Aware Feature Encoder**: Transforms RGB images into 3D occupancy features using Voxel-Image Attention and Cross-Voxel Attention
- **Altitude-Aware Feature Reduction**: Compresses 3D occupancy along the altitude dimension to 2D, reducing transmission cost by 97% (from 76.56 MB to 0.23 MB per message)
- **Cross-Agent Feature Integration**: Fuses compressed features from multiple UAVs using attention mechanisms weighted by view overlap
- **Dual-Mask Perceptual Guidance**: Adaptively selects informative features based on scene complexity and UAV viewpoint

Experimental Results: Evaluated on three datasets (Air-to-Pred-Occ, UAV3D-Occ, GauUScene-Occ), MCOP achieved:

- **mIoU**: 47.17 (collaborative), 30.24 (standalone) on UAV3D-Occ
- **Communication Volume**: 0.23 MB per transmission vs. 76.56 MB for uncompressed 3D features
- **Perception Range**: 500×500m at 150m altitude, far exceeding single-UAV capabilities

MCOP demonstrates that intelligent compression and selective feature sharing enable large-scale UAV swarms to perform complex 3D perception tasks without overwhelming network infrastructure, critical for applications in disaster response, search and rescue, and environmental monitoring.

6.3.15. Multi-Robot SLAM in Dynamic Networks

Robust localization and mapping under intermittent connectivity and dynamic network topologies remain critical for real-world deployments. Agizza and Isler proposed monocular SLAM utilizing rendezvous [2], where robots operating independently periodically meet at designated locations to exchange map information and perform loop closure. Unlike continuous communication approaches, rendezvous-based coordination reduces bandwidth requirements while still enabling global consistency. Experiments with ground robots in indoor environments demonstrated comparable mapping accuracy to fully connected systems with 70% less communication overhead.

Tian et al. developed asynchronous and parallel distributed pose graph optimization [52] for multi-robot SLAM back-ends. Their algorithm allows robots to optimize local pose graphs independently while asynchronously incorporating information from neighbors, avoiding the need for global synchronization barriers. Convergence guarantees hold under mild connectivity assumptions, and experiments with 10+ robot teams showed near-linear speedup compared to centralized bundle adjustment. This work is particularly relevant for large-scale deployments where centralized optimization becomes infeasible.

The resilience of multi-robot SLAM to sensor failures, loop closure errors, and adversarial attacks has been studied extensively. Tian et al. released comprehensive datasets and benchmarks for resilient distributed visual SLAM [51], including scenarios with outlier loop closures, intermittent GPS, and Byzantine agents intentionally corrupting shared information. Their evaluation of distributed SLAM algorithms revealed that most methods degrade gracefully under moderate failure rates but catastrophically fail when Byzantine agents exceed 20-30% of the team. This work highlights the need for robust outlier rejection and redundancy mechanisms in safety-critical applications.

6.3.16. Specialized Applications: Warehousing, Agriculture, and Off-Road Navigation

Warehouse automation has driven significant advances in multi-robot path planning under tight space constraints. Huo et al. proposed path planning in narrow warehouse environments with fast feasibility heuristics [26]. Their approach pre-computes feasibility zones for different robot

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

configurations, enabling real-time collision checking and path refinement as robots navigate through aisles with minimal clearance. Experiments in simulated and real warehouse environments demonstrated 50% faster planning compared to sampling-based methods while maintaining solution quality.

Off-road navigation introduces uncertainty in terrain traversability and robot dynamics. Cai et al. developed risk-aware navigation via learned speed distribution maps [8], where robots estimate traversability not as binary (traversable/non-traversable) but as distributions over achievable speeds. Multi-robot teams share traversability observations to collaboratively build speed maps, improving navigation efficiency and safety. Field tests with ground robots in outdoor environments showed that risk-aware planning reduced navigation failures by 40% compared to geometric obstacle avoidance alone.

6.3.17. Communication and Learning in Multi-Agent Systems

Learning when and what to communicate represents a significant challenge in multi-agent reinforcement learning. Fang et al. proposed learning to communicate and correct pose errors [18], where agents discover communication strategies that mitigate localization drift. Their approach uses learned attention mechanisms to selectively share observations that maximally reduce team-wide pose uncertainty. Experiments with quadrotors demonstrated that learned communication outperformed fixed-schedule sharing, reducing localization error by 35% while using 60% less bandwidth.

Spectrum allocation for Device-to-Device (D2D) communication among mobile robots has been addressed through Multi-Agent Reinforcement Learning (MARL). Farooq and Imran proposed spectrum allocation for D2D underlay communications [20], where robots learn to dynamically allocate frequency bands to maximize throughput while minimizing interference. Their MARL approach outperformed heuristic allocation schemes in dense robot deployments, demonstrating the value of learning-based methods for managing radio resources in multi-robot systems.

6.3.18. Dense Scene Reconstruction and Coverage Planning

Collaborative 3D reconstruction benefits from coordinated viewpoint selection. Dong et al. developed multi-robot dense scene reconstruction [16] using neural implicit representations and active view planning. Their system optimizes camera trajectories to minimize reconstruction uncertainty while respecting inter-robot collision constraints. Experiments with quadrotors reconstructing complex 3D structures showed 25% fewer required views compared to independent coverage planning, enabled by learned models predicting information gain from potential viewpoints.

Coverage path planning with optimality guarantees has been studied using spanning tree methods. Tang et al. proposed ideal-shaped spanning tree coverage [50], where robots collaboratively construct a spanning tree over the workspace with bounded detour from the optimal Euclidean Steiner tree. Their distributed algorithm ensures complete coverage while minimizing overlap and repeated traversals, validated through experiments with ground robots in indoor environments.

6.3.19. Temporal Logic Specifications and Formal Methods

Complex mission requirements are increasingly expressed using temporal logic, enabling formal verification and automated synthesis. Bai et al. addressed task planning under individual and collaborative temporal logic specifications [3], where each robot has local goals (e.g., "avoid region A until sensing region B") and the team has shared objectives (e.g., "all robots must visit region C within 10 time steps"). Their synthesis algorithm generates provably correct motion plans by solving a graph search problem over the product automaton of individual and collaborative specifications. Experiments with mobile robots demonstrated successful execution of complex missions that would be intractable to specify through traditional reward engineering.

6.3.20. Graph Neural Networks for Decentralized Action Selection

Submodular optimization appears naturally in multi-robot coverage, exploration, and target tracking. Zhou and Tokekar proposed GNN-based decentralized action selection [62], where robots learn to greedily maximize submodular objectives without explicit coordination or centralized decision-making. Their Graph Neural Network architecture enables each robot to make locally optimal decisions that provably achieve near-global optimality through message passing with neighbors. Experiments in exploration and target tracking scenarios showed performance within 10% of centralized greedy selection while operating fully decentralized.

Extending GNN-based decentralized reasoning to the task assignment domain, **DGNN-GA** [22] demonstrates that heterogeneous graph neural networks can solve linear sum assignment problems without centralized computation. Unlike the submodular action selection approach above, which maximizes coverage objectives, DGNN-GA addresses the combinatorial assignment of robots to specific goal locations by constructing a bipartite heterogeneous graph with robot and goal node types. Edge features encode distance and capability information, and the GNN learns to classify edges as active (assigned) or inactive through local message passing. This formulation naturally handles unequal robot-to-goal ratios and scales to 50 robots with zero-shot transfer from smaller training instances, offering a complementary GNN-based tool for multi-robot coordination alongside coverage-oriented methods.

6.3.21. Distributed Graph Optimization for Relative Localization

Latif and Parasuraman introduced DGORL (Distributed Graph Optimization-based Relative Localization) [31], a decentralized approach to multi-robot relative pose estimation. Unlike centralized bundle adjustment, DGORL distributes optimization across robots using consensus algorithms, where each robot maintains a local estimate of the global pose graph

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

and iteratively refines it through information exchange with neighbors. The method handles dynamic network topologies and asynchronous communication, making it robust to connectivity changes common in real deployments. Experiments with ground and aerial robots demonstrated convergence to accurate relative poses even under 30% packet loss, highlighting its suitability for bandwidth-limited or unreliable networks.

6.3.22. Underground and GPS-Denied Exploration

Extreme subterranean environments present unique challenges including perceptual degradation from darkness and dust, self-similar repetitive structures causing aliasing, and severe communication constraints from multi-path interference and range limitations. **LAMP 2.0** [11] was specifically designed for such scenarios, integrating robust LiDAR-based SLAM with multi-stage loop closure detection and outlier-resilient optimization.

The system architecture separates concerns across multiple layers: a sensor-agnostic front-end interface supports diverse odometry sources (Visual-Inertial Odometry (VIO), wheel odometry, Light Detection and Ranging (LiDAR) odometry) and LiDAR configurations (spinning, solid-state, multi-beam), enabling deployment on heterogeneous platforms. The multi-robot front-end performs both intra-robot loop closure (within a single robot's trajectory) and inter-robot loop closure (between different robots) using SC++ scan context descriptors for initial place recognition followed by geometric verification through Iterative Closest Point (ICP) registration.

The back-end employs a robust pose graph optimizer based on graduated non-convexity, which progressively filters outlier loop closures by gradually transitioning from a convex approximation to the original non-convex optimization problem. This approach successfully rejects false positives that would otherwise corrupt the map in perceptually-degraded environments. Extensive validation during the DARPA Subterranean Challenge Finals demonstrated LAMP 2.0's effectiveness across a heterogeneous team of 4 UAVs, 6 ground robots, and 2 legged robots exploring over 6 kilometers of underground mines, tunnels, and cave networks. The system maintained consistent localization accuracy below 2 meters across the entire deployment, enabling artifact detection, survivor localization, and safe autonomous navigation in one of the most challenging robotics competitions to date.

6.3.23. Opportunistic Multi-Robot Exploration

Large-scale outdoor exploration scenarios often involve terrain that blocks line-of-sight communication, creating intermittent connectivity where traditional coordination fails. **MOCHA** [39] addresses these challenges through opportunistic communication protocols that require only brief, sporadic connectivity windows to maintain team coordination.

The key insight is epidemic-style information dissemination: when two robots come within communication range, they perform bidirectional synchronization exchanging new map segments, detected artifacts, and exploration frontiers using compressed representations. Each robot maintains a local belief over the global state, which is incrementally updated through gossip exchanges. Critical information (e.g., safety hazards, mission-critical artifacts) is prioritized for immediate transmission, while routine updates (e.g., marginal map refinements) are transmitted when bandwidth permits.

MOCHA coordinates heterogeneous teams where UAVs exploit aerial mobility to serve as mobile relay nodes, periodically descending to exchange information with ground robots that may be radio-isolated in valleys or dense forest. Field experiments in forest environments demonstrated that with UAV-assisted opportunistic communication, ground robots isolated for up to 5 minutes could still contribute to team-wide situational awareness through eventual information propagation. The system scaled to 12 heterogeneous platforms (8 ground robots, 4 UAVs) exploring over 2 square kilometers with sustained coordination despite network fragmentation and 50% average link loss rates.

6.3.24. Active SLAM and Coordinated Exploration

Traditional frontier-based exploration assigns robots to nearest unexplored regions, often resulting in redundant coverage and suboptimal information gain. **Efficient Multi-robot Active SLAM** [19] integrates uncertainty-aware exploration planning with collaborative SLAM to maximize both coverage efficiency and localization accuracy.

The system's utility function combines three factors: frontier path entropy (measuring map uncertainty reduction), travel distance cost, and merged map information gain (quantifying expected improvement from inter-robot loop closures). Frontiers are extracted from each robot's occupancy grid and scored according to this utility function, with assignments coordinated through a centralized server running Hungarian algorithm-based matching. To prevent low-priority robots from being starved when high-value frontiers are distant, the system implements dynamic priority escalation where repeatedly preempted robots receive temporary priority boosts.

The collaborative SLAM component continuously fuses individual robot maps using Oriented FAST and Rotated BRIEF (ORB) feature-based place recognition, triggering global pose graph optimization when inter-robot loop closures are detected. This tight coupling between exploration planning and SLAM refinement ensures that robots not only cover space efficiently but also reduce localization uncertainty through coordinated revisits to previously mapped regions.

Validation through both Gazebo simulations (with multiple TurtleBots in office environments) and real-world experiments demonstrated 20-30% faster exploration compared to greedy frontier selection, with 40% reduction in total travel distance and improved final map quality (measured by feature density and loop closure consistency). The system particularly excels in structured environments where revisiting regions from different viewpoints provides high information gain for SLAM accuracy.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

6.3.25. *Reinforcement Learning for Dynamic Task Allocation*

Warehouse and logistics operations present complex task allocation challenges including varying object weights, infeasible tasks that exceed single-robot capabilities, and dynamic task arrivals requiring real-time reallocation. Traditional optimization-based methods struggle with these uncertainties and dynamic constraints.

Suzuki et al.'s **RL-based multi-robot task allocation** [48] explicitly models task feasibility within the reinforcement learning framework. The state representation includes robot positions, capabilities (lifting capacity, manipulator reach), object locations, object properties (weight, fragility), and a feasibility map indicating which robots can handle which objects. The action space encompasses both task assignment and infeasible task detection, where the agent learns to identify when an object requires cooperative manipulation or should be flagged for human intervention.

The reward function penalizes failed pick attempts, incentivizes successful deliveries, and provides negative rewards for deadlock states where robots indefinitely wait for impossible tasks. Training through Deep Q-Networks in simulated warehouse environments with randomized object distributions, the system learned policies that outperformed fixed allocation heuristics by 25-30% in task throughput. Real-world deployment on Fetch mobile manipulators validated the sim-to-real transfer, with the RL policy successfully detecting and reporting infeasible heavy objects while efficiently allocating feasible tasks.

Extending beyond reactive allocation, Ye et al.'s **two-stage RL framework** [59] decomposes resource allocation into strategic activation and tactical assignment. In power grid scenarios, the first-stage RL agent decides which backup generators to activate based on forecasted demand, balancing startup costs against expected load. The second-stage agent then assigns power transmission tasks to activated resources through graph-based optimization. This hierarchical decomposition reduces the combinatorial action space from exponential to polynomial complexity while enabling explicit cost-benefit reasoning about resource activation.

Experiments across power grid management and logistics domains demonstrated that two-stage RL outperformed end-to-end RL baselines by 15-20% in total operational cost, primarily by avoiding unnecessary resource activations when demand could be met with already-active resources. The approach generalizes to multi-robot scenarios where activating robots incurs setup overhead (battery swapping, initialization), making selective activation crucial for energy efficiency.

6.4. Learning Paradigms for Collaboration

Learning augments handcrafted coordination rules with data-driven adaptability. Wu and Suh's survey [54] categorizes robot learning approaches into supervised, reinforcement, imitation, transfer, causal inference, ensemble, and meta-learning.

Supervised learning is commonly used for perception modules, such as object detection and semantic segmentation, supporting shared situational awareness.

Multi-agent reinforcement learning (MARL) enables robots to learn cooperative strategies but is often sample-inefficient and sensitive to the sim-to-real gap.

A critical dimension of MARL that has received increasing attention is *heterogeneity*. Bettini et al. introduced **HetGPPO** (Heterogeneous Graph-based Proximal Policy Optimization), a framework for learning truly heterogeneous multi-agent policies with GNN-based inter-agent communication [6]. The authors propose a taxonomy of multi-robot heterogeneity: *physical heterogeneity* (different dynamics, sensors, or actuators), *behavioral heterogeneity with same objectives* (different strategies toward a common goal), and *behavioral heterogeneity with different objectives* (distinct sub-goals within a shared mission). HetGPPO removes parameter sharing between agents, allowing each robot to learn an independent policy while still communicating through a shared GNN communication channel. A key finding is that while homogeneous behavioral typing (where agents share a single policy) performs well in idealized simulation, it becomes brittle under real-world noise and perturbations. Truly heterogeneous policies learned by HetGPPO demonstrate superior robustness in sim-to-real transfer scenarios, as each agent can specialize its behavior to compensate for individual sensor noise and actuator variations. However, this comes at the cost of reduced sample efficiency during training due to the larger parameter space.

Distributed learning frameworks such as **DiNNO** overcome data centralization limitations by allowing each robot to train locally and share model weights with neighbors [60]. DiNNO converges nearly as fast as centralized training and outperforms other distributed methods like DSGD and DSGT, even under non-iid data distributions.

Imitation and transfer learning accelerate policy convergence, while **meta-learning** improves adaptation to novel tasks with minimal additional data.

Language-driven learning, exemplified by RoCo and COHERENT, introduces interpretable reasoning and human-in-the-loop collaboration, marking a shift toward explainable multi-robot systems.

6.5. Communication-Efficient Decentralized Learning

Decentralized learning frameworks continue to evolve to reduce communication load while preserving learning performance across heterogeneous robot networks. Huang et al. proposed **CoCoL** (Communication-Efficient Decentralized Collaborative Learning), a framework that addresses high bandwidth consumption and non-identical data distributions across robots [25].

CoCoL combines *mirror descent optimization* with *gradient tracking* and approximate Newton-type updates, allowing each robot to train locally while synchronizing only essential information. By exploiting the similarity between robots'

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

objective functions, the system approximates curvature information without expensive Hessian computation. Each robot performs several local Adam updates before communicating model parameters, striking a balance between accuracy and efficiency.

Evaluations across three multi-robot learning tasks that are **image classification**, **neural implicit mapping**, and **multi-agent reinforcement learning** showed that CoCoL reduces bandwidth usage by up to 53% compared to DiNNO and improves SSIM in neural mapping by 33%. It also achieved near-centralized accuracy in reinforcement learning while converging faster than distributed gradient tracking (DSGT) and SONATA. Although CoCoL currently lacks formal convergence proofs under stochastic settings, it establishes a strong foundation for scalable, communication-efficient learning, particularly for resource-constrained robotic swarms.

6.6. Collaborative Manipulation and Ability-Aware Control

Yan et al. introduced **Decentralized Ability-Aware Adaptive Control (DA3C)**, which accounts for each robot's force capability during collaborative manipulation [55]. DA3C aligns robot configurations to maximize force in the desired task direction, adapts control gains online, and shares minimal "deficiency signals" when a robot approaches saturation. Experiments show improved tracking accuracy and robustness to mass and friction changes, making it a promising approach for cooperative transportation and manipulation tasks.

Overall, these works highlight an ecosystem of methods that collectively advance the scalability, interpretability, and robustness of multi-robot systems.

7. Comparative Analysis

A comparative summary of the reviewed works is presented in Table 3, highlighting trade-offs across communication architecture, scalability, interpretability, and real-world deployment. Centralized planners achieve globally optimal task assignments but suffer from single points of failure and poor scalability, while decentralized frameworks such as Swarm-SLAM and DA3C improve resilience but require sophisticated consensus algorithms. Hybrid systems like HMAS and COHERENT balance centralized guidance with local autonomy. Recent advances span diverse coordination strategies: agent-server architectures offload computation to powerful servers while maintaining lightweight onboard control; concurrent transmission achieves microsecond-level synchronization at increased complexity cost; protocol-aware systems reveal dramatic communication load variations (LoRa for agriculture, Wi-Fi for warehouses); game-theoretic Control Barrier Functions provide safety guarantees but face scalability limits; neural bipartite matching outperforms greedy baselines at high training cost; homomorphic encryption addresses security at significant computational overhead. Multi-human multi-robot interfaces require both direct verbal and indirect visual communication for operator scalability. Collaborative perception highlights centralized fusion accuracy versus distributed robustness trade-offs. Learning-based systems like YOLOv5-PPO-A3C achieve scalability through asynchronous updates. Distributed localization methods trade sensing requirements for convergence speed: range-based approaches eliminate GPS dependency but converge slowly; rendezvous-based SLAM reduces communication overhead 70% through periodic exchange; asynchronous pose graph optimization achieves near-linear speedup. UAV-specific compression, formal temporal logic methods (limited to 5-10 robots by state explosion), and application-specific optimizations (warehouse aisles, off-road terrain, environmental sampling) demonstrate that domain-tailored approaches outperform generic algorithms. Overall, persistent tensions exist between theoretical optimality, computational tractability, communication efficiency, and real-world robustness, with laboratory success often failing under field deployment conditions due to sensor noise, network unreliability, and environmental variability.

Recent distributed SLAM frameworks demonstrate high scalability and low communication overhead through selective information sharing. Swarm-SLAM's spectral prioritization and SlideSLAM's semantic representations achieve scalability rating of 5 while maintaining communication load at level 2, validated through extensive field testing including the DARPA Subterranean Challenge. Multi S-Graphs [21] pushes communication efficiency further by exchanging hierarchical semantic-relational graphs rather than geometric data, achieving 95% data reduction with improved localization accuracy. LAMP 2.0 represents the current state-of-the-art for extreme environments, successfully coordinating 12+ robots across 6 km of underground terrain with sustained localization accuracy. MOCHA pushes the boundaries of intermittent connectivity tolerance, achieving very high scalability (5) with minimal communication requirements (1) through epidemic-style information dissemination. Task-driven map compression [42] achieves the lowest communication load (1) through intelligent abstraction template selection, demonstrating 98% data reduction for navigation tasks. GNN-based methods such as DGNN-GA [22] achieve high scalability for decentralized task assignment with minimal communication rounds, while HetGPPO [6] demonstrates that heterogeneous MARL policies can improve robustness under real-world conditions at moderate communication cost. Active SLAM frameworks balance exploration efficiency with coordination overhead, while RL-based task allocation methods demonstrate adaptability to dynamic constraints at moderate communication costs. Distributed variational inference for object SLAM [9] achieves moderate scalability with probabilistic uncertainty quantification, though evaluated primarily in simulation environments.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Table 3: Comparison of Representative Multi-Robot Collaboration Approaches

Approach	Scalability (1-5)	Comm. Load (1-5)	Interpretability (1-5)	Real-World Validation
Centralized Planner	2	5	3	Simulation
Decentralized Consensus	5	3	2	Limited
Swarm-SLAM	5	2	3	Real Robots [Lab]
HMAS	5	3	4	Simulation
COHERENT	5	3	4	Sim + Real
RoCo	4	3	5	Simulation
HMCF	5	3	5	Sim + Real
Omnid Mocobots	3	1	4	Real Robots [Manipulation]
DA3C Control	3	1	3	Simulation
Wireless Collaboration	3	3	3	Semi-physical
ROS Disaster Exploration	3	3	3	Real Robots [Disaster]
Aerial-Ground Construction	3	2	4	Simulation
ACHORD	4	2	3	Real Robots [Field Tested]
Online Learning + Connectivity	4	2	2	Simulation
CoSPAM	4	3	3	Simulation
CoCoL	5	1	3	Simulation
Concurrent Transmission	4	1	3	Real Robots [5 units]
Protocol Analysis	5	2-4*	4	HPC Simulation
MHMRI	3	3	5	User Study [18 participants]
Collaborative Perception	4	4	4	Simulation
YOLOv5-PPO-A3C	5	3	4	Multi-dataset
MCOP	5	1	3	Dataset Validation [3]
MR-DTG	5	2	3	Real Robots [6 UAVs]
Agent-Server Architecture	4	3	4	Photo-realistic Sim
Game-Theoretic CBF	3	2	4	Real Robots [Navigation]
Neural Bipartite Matching	4	3	3	Simulation
Range-Based Localization	5	2	3	Real Robots [GPS-denied]
Task Alloc. w/ Range Constraints	3	3	4	Simulation
Heterogeneous Water Sampling	4	3	4	Real Robots [Field Trials]
Macro-Action Q-Learning	4	2	3	Real Robots [Warehouse]
Homomorphic Encryption	2	4	2	Simulation
Target Tracking w/ Danger Zones	4	3	3	Simulation
Rendezvous-Based SLAM	4	2	3	Real Robots [Indoor]
Risk-Aware Off-Road Nav	3	2	3	Real Robots [Outdoor]
Resilient Distributed SLAM	5	3	3	Dataset Released
Narrow Warehouse Planning	3	2	4	Sim + Real [Warehouse]
GNN Action Selection	5	2	3	Simulation
Learning Pose Correction	4	2	3	Real Robots [Quadrotors]
Spanning Tree Coverage	4	2	4	Real Robots [Indoor]
Async Pose Graph Optim	5	2	3	Real Robots [10+ teams]
MARL Spectrum Allocation	5	3	2	Simulation
Dense Scene Reconstruction	3	4	3	Real Robots [Quadrotors]
Temporal Logic Planning	3	2	5	Real Robots [Mobile]
DGORL Localization	5	2	3	Real Robots [Dynamic Net]
Swarm-SLAM	5	2	3	Real Robots [Field Tested]
SlideSLAM	5	2	4	Real Robots [Multi-platform]
LAMP 2.0	5	2	3	Real Robots [DARPA SubT]

Continued on next page

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Approach	Scalability (1-5)	Comm. Load (1-5)	Interpretability (1-5)	Real-World Validation
MOCHA	5	1	3	Real Robots [Field Tested]
Efficient Multi-robot A-SLAM	4	3	3	Sim + Real
RL Task Allocation	4	2	3	Sim + Real
Two-Stage RL Allocation	4	2	3	Simulation
Multi-Robot Object SLAM	4	3	3	Simulation
Multi S-Graphs	5	1	4	Sim + Real [Multi-floor]
DGNN-GA Goal Assignment	5	2	3	Sim + Real [Quadrotors]
HetGPPO	4	2	3	Simulation [Multi-scenario]
Comm-Aware Map Compression	4	1	3	Simulation [Mars/Earth]

Scale: 1 = Very Low, 2 = Low, 3 = Moderate, 4 = High, 5 = Very High.

*Protocol Analysis varies by protocol: LoRa (2), Zigbee (2), Bluetooth (3), Wi-Fi (4)

Table 4: Comparative Summary of Multi-Robot Coordination Paradigms

Criterion	Classical (Rule-Based)	Optimization-Based	RL-Based	LLM-Driven	Key Trade-offs
Scalability	Moderate (10–50 robots); degrades with team size	Limited (5–20 robots); NP-hard complexity	High (50–100+ robots); parallelizable training	Moderate (5–30 robots); token limits constrain team size	RL scales best; optimization worst
Interpretability	High; explicit rules easily auditable	Moderate; objective functions interpretable, solutions less so	Low; black-box policies hard to explain	Very High; natural language reasoning transparent	LLM excels; RL struggles
Deployment Readiness	High; mature, field-proven systems	Moderate; requires parameter tuning per environment	Low–Moderate; sim-to-real gap significant	Low; computational cost, latency, hallucination risks	Classical most mature; LLM least
Adaptability	Low; manual rule updates needed	Low–Moderate; re-optimization required for new scenarios	High; learns from experience	High; few-shot adaptation via prompting	RL and LLM adapt; classical/optim. rigid
Communication Load	Low–Moderate; predefined protocols	Moderate–High; iterative optimization messages	Variable; centralized training, decentralized execution	Moderate–High; language tokens exchanged	Classical lowest; LLM/optim. higher
Computation Cost	Low; lightweight onboard	High; solver complexity grows exponentially	High (training); Low–Moderate (inference)	Very High; LLM inference dominates	Classical cheapest; LLM most expensive
Robustness to Failures	Moderate; handcrafted fallbacks	Low; single-point optimization failures	Moderate; policy generalization varies	Moderate; degrades gracefully with prompting	Decentralized RL most robust
Human Integration	Low; rigid command interfaces	Low; indirect via objective specification	Low–Moderate; reward shaping from feedback	Very High; natural language interaction	LLM enables intuitive HRI
Representative Systems	Consensus protocols, Behavior trees, FSMs	MILP planners, Hungarian algorithm, CBF methods	MARL, PPO-based coordination, Q-learning allocation, DGNN-GA, HetGPPO	RoCo, HMCF, COHERENT, Co-NavGPT	–

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

To provide a clearer perspective on the evolution of multi-robot coordination paradigms, Table 4 presents a comparative summary contrasting classical, optimization-based, reinforcement learning-based, and LLM-driven approaches across scalability, interpretability, and deployment readiness. Classical rule-based methods offer predictable behavior and straightforward deployment but lack adaptability. Optimization-based approaches provide theoretical guarantees but face computational bottlenecks at scale. RL-based methods excel at learning adaptive policies but suffer from sample inefficiency and sim-to-real gaps. LLM-driven coordination represents the emerging frontier, offering unprecedented interpretability through natural language reasoning, though deployment readiness remains limited by computational requirements and hallucination concerns.

8. Open Challenges

Despite significant progress, several challenges remain before large-scale, robust multi-robot systems can be deployed reliably in unstructured real-world environments. We organize these challenges into thematic categories reflecting the multi-faceted nature of multi-robot collaboration.

8.1. Communication and Networking Challenges

Scalable Communication Beyond Hundreds of Robots: Even with sparse communication frameworks like Swarm-SLAM and compressed features in MCOP, coordinating hundreds to thousands of robots with minimal bandwidth remains unsolved. Network partitioning, asynchrony, and the combinatorial explosion of potential inter-robot interactions introduce complexity that current methods struggle to address.

Task-driven compression frameworks like [42] demonstrate that up to 98% data reduction is achievable for 2D grid-based maps, but extending such approaches to 3D volumetric representations, dynamic environments with moving obstacles, and heterogeneous map formats (semantic maps, topological graphs, occupancy grids) remains an open challenge. Moreover, jointly optimizing compression strategy, exploration behavior, and communication scheduling presents a coupled optimization problem that current methods address only in isolation.

Intermittent Connectivity and Information Aging: Opportunistic communication frameworks like MOCHA tolerate network partitions through epidemic-style dissemination, but information aging introduces subtle challenges. When robots operate disconnected for extended periods (minutes to hours), their local beliefs about the global state become stale. Upon reconnecting, reconciling conflicting beliefs (e.g., two robots independently claiming to have explored the same region) requires sophisticated conflict resolution beyond simple timestamp-based arbitration. Byzantine agreement protocols could address intentional misinformation, but their message complexity may be prohibitive for bandwidth-limited scenarios. Developing lightweight, eventually-consistent information fusion protocols that gracefully handle both benign staleness and adversarial corruption is critical for long-duration missions.

Gossip Protocol Overhead in Dense Networks: MOCHA's epidemic dissemination excels in sparse, intermittently connected networks, but may introduce excessive redundancy in densely connected scenarios where many robots simultaneously share overlapping information. Adaptive gossip rates that scale inversely with network density, or probabilistic suppression mechanisms where robots randomly defer transmission if they detect neighbors already broadcasting, could reduce redundant traffic. However, ensuring critical information still propagates with high probability requires careful tuning of suppression parameters.

8.1.1. Distributed SLAM Scalability and Consistency

While Swarm-SLAM demonstrates 60% communication reduction through spectral prioritization, fundamental limits remain on how large teams can scale before loop closure verification overhead dominates computation budgets. Current systems validate inter-robot loop closures through geometric consistency checks (e.g., ICP registration), which scale quadratically with the number of candidate correspondences. For teams exceeding 50-100 robots, even with spectral filtering, the sheer volume of inter-robot loop closure candidates becomes intractable.

Moreover, ensuring global map consistency in fully decentralized systems without any centralized coordination remains an open problem. Swarm-SLAM and SlideSLAM employ decentralized pose graph optimization, but convergence guarantees weaken under extreme network partitions or Byzantine robots intentionally corrupting shared loop closures. Developing robust distributed consensus mechanisms that maintain map consistency while tolerating up to 30-40% malicious agents requires further research, particularly for safety-critical applications where adversarial tampering is a concern.

Semantic SLAM approaches like SlideSLAM offer viewpoint-invariant place recognition, but their robustness degrades in feature-sparse environments (empty corridors, natural outdoor settings with minimal objects). Probabilistic approaches using distributed variational inference [9] provide principled uncertainty quantification but face scalability challenges when maintaining full covariance estimates across large teams. Hybrid systems combining geometric and semantic representations could improve robustness, but efficiently fusing these modalities while maintaining real-time performance on resource-constrained platforms remains challenging.

Protocol Adaptability and Hybrid Networking: Comparative protocol analysis reveals no universal solution; each protocol exhibits distinct trade-offs between range, throughput, latency, and power consumption. Future systems must dynamically switch protocols based on real-time conditions (e.g., transitioning from LoRa for long-range updates to Zigbee for close-range coordination during clustering). Hybrid protocol stacks that seamlessly integrate multiple technologies remain largely unexplored.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Concurrent Transmission in Heterogeneous Teams: While split architecture approaches demonstrate concurrent transmission viability in homogeneous robot teams, extending this paradigm to heterogeneous platforms with varying computational capabilities, sensor suites, and communication hardware poses significant synchronization challenges. Adaptive time-slot allocation and priority-based transmission scheduling are needed to accommodate processing speed differences without degrading network performance.

Network Reliability and Latency: Wireless and ROS-based frameworks remain vulnerable to bandwidth drops and network partitioning, which can degrade collaborative SLAM and coordination performance. Coordinating motion using online communication estimates requires adaptive control strategies that remain reliable under fast signal variations and noisy link-quality learning models. Agent-server architectures partially address this by concentrating computation at a central server, but introduce new failure modes when server connectivity is lost.

Privacy and Security in Distributed Coordination: Homomorphic encryption enables privacy-preserving coordination but introduces prohibitive computational overhead (10-100× slowdown for encrypted path planning). Lightweight cryptographic methods that balance security with real-time performance are needed. Additionally, Byzantine-robust algorithms that tolerate adversarial robots are critical for safety-critical applications, yet current methods fail when malicious agents exceed 20-30% of the team. Distributed consensus mechanisms must detect and isolate faulty or compromised robots without excessive communication overhead.

8.2. Human-System Interaction Challenges

Scalability of Human Supervision: Multi-human multi-robot interaction studies show that mixed communication enhances dual-operator coordination, but scaling to larger operator teams (10, 50, or 100 operators overseeing thousands of robots) introduces unresolved questions. Hierarchical interfaces, attention management systems, and automated situation summarization are necessary but underdeveloped.

Human-in-the-Loop Stability and Interpretability: Frameworks that integrate human feedback and natural-language reasoning into robot coordination must ensure stable interaction while preventing over-dependence on human input. Future systems must balance autonomy with human oversight while preserving transparency, trust, and task efficiency.

Calibrating Trust and Automation Reliance: Determining optimal human intervention frequency remains critical. Excessive involvement reduces automation benefits, while insufficient oversight risks undetected failures. Adaptive autonomy frameworks that calibrate intervention thresholds based on operator expertise and task criticality are needed.

8.3. Perception and Learning Challenges

Centralized vs. Distributed Perception: Collaborative perception studies demonstrate performance gains from centralized fusion but introduce single points of failure and bandwidth bottlenecks. Distributed fusion algorithms that maintain global consistency without centralized aggregation (e.g., gossip-based averaging, blockchain-verified consensus) need further development. Agent-server models offer a middle ground by centralizing computation without requiring all robots to share raw sensor data, yet balancing server load with team size remains challenging.

Hierarchical semantic-relational approaches like Multi S-Graphs [21] address communication efficiency through structured abstractions (rooms, walls, floors), achieving over 95% data reduction. However, such methods depend on environments with identifiable structural elements, limiting applicability in unstructured outdoor settings (forests, open terrain) where room-level abstractions do not exist. Developing adaptive hierarchical representations that automatically discover appropriate abstraction levels for diverse environments remains an open problem.

Sim-to-Real Transfer: Multi-agent reinforcement learning policies trained in simulation often fail to generalize to the real world due to sensor noise, unmodeled dynamics, and distribution shifts. Domain randomization, meta-learning, and real-world data bootstrapping are still areas of active research. Macro-action learning and learned communication strategies show promise in reducing training complexity but require extensive simulation time that may not transfer to hardware.

Multi-Modal Learning Integration: Integrating visual, semantic, and sensor modalities into unified representations for decision-making is non-trivial. Current approaches require extensive compute resources and labeled datasets, limiting practical deployment. Attention mechanisms, transformer architectures, and contrastive learning offer potential solutions but remain computationally expensive.

Learning When and What to Communicate: Multi-agent reinforcement learning for communication strategies shows promise but requires extensive simulation training that often fails to transfer to real hardware. Developing meta-learning approaches that quickly adapt communication policies to new environments and robot teams remains an open problem. Current learned communication methods also struggle with bandwidth variability and packet loss common in real wireless networks.

Relative Localization Without Infrastructure: Range-only localization avoids dependence on GPS or visual landmarks but suffers from slower convergence and ambiguity in certain geometric configurations. Hybrid methods combining minimal sensing modalities (range, IMU, occasional visual features) could provide robust localization in diverse environments. Rendezvous-based approaches reduce communication requirements but introduce coordination overhead for scheduling meeting locations.

Semantic Representation Ambiguity: Object-level SLAM like SlideSLAM assumes distinct, recognizable landmarks, but many real-world environments contain repeated or ambiguous objects (e.g., rows of identical shelves in warehouses, uniformly spaced trees in forests). Data association ambiguity, i.e., determining whether two robots observe the

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

same object or two identical objects, remains a fundamental challenge. Probabilistic data association incorporating spatial priors and multi-hypothesis tracking could improve robustness, but increases computational complexity. Current systems often rely on geometric verification as a final arbiter, limiting the benefits of semantic abstraction when geometric appearance is also ambiguous (e.g., featureless walls).

Active SLAM with Multi-Objective Constraints: Efficient Multi-robot Active SLAM balances exploration efficiency with localization accuracy, but real deployments introduce additional objectives: energy consumption (longer paths drain batteries faster), communication maintenance (exploration must preserve network connectivity), and safety (avoiding hazardous regions). Multi-objective optimization frameworks that jointly reason about these competing criteria remain underdeveloped. Pareto-optimal frontier selection could enable operators to tune exploration behavior along a coverage-accuracy-safety trade-off curve, but computing Pareto frontiers in real-time for large frontier sets poses computational challenges.

8.4. Safety and Robustness Challenges

Safety Under Adversarial Conditions: LLM-driven planners must ensure safety under adversarial inputs, sensor errors, or out-of-distribution scenarios. Runtime verification and fail-safe switching between policies are required to prevent unsafe behaviors from language model hallucinations or misinterpreted commands. Game-theoretic approaches provide formal safety guarantees through Control Barrier Functions but struggle to scale beyond small teams due to computational complexity of Nash equilibrium computation.

Physical and Mechanical Communication: The emergence of mechanically coupled systems introduces new challenges in force-based intent sharing, compliance control, and co-manipulation safety. Developing unified control models that combine physical interaction cues with networked communication could improve the robustness of hybrid human-robot and robot-robot collaboration.

Sensor Errors and Occlusions: While collaborative perception improves robustness, it does not eliminate failures. Redundant sensing modalities, Bayesian fusion, and uncertainty quantification should be integrated to handle sensor degradation gracefully. Resilient SLAM systems must detect and reject outlier loop closures while tolerating Byzantine agents that intentionally corrupt shared map information.

8.5. Heterogeneity and Interoperability Challenges

Heterogeneous Teaming: Demonstrations with mixed teams show promise, but scaling to more diverse robots (UAVs, quadrupeds, manipulators, marine vehicles) under different dynamics and sensing modalities is an open problem. Software incompatibilities (ROS1 vs. ROS2, proprietary SDKs), hardware differences (communication standards, power systems), and capability mismatches complicate coordination. Heterogeneous systems for environmental monitoring demonstrate 40% efficiency improvements but require manual tuning of task allocation strategies for each robot type.

At the policy level, HetGPPO [6] demonstrates that allowing each robot to learn an independent policy (rather than sharing parameters across agents) improves robustness under real-world noise and perturbations. However, removing parameter sharing significantly increases the number of trainable parameters, reducing sample efficiency and increasing training time. Determining the optimal degree of policy heterogeneity—fully shared, partially shared (e.g., shared perception layers with independent action heads), or fully independent—remains an open challenge that depends on team composition, task structure, and deployment conditions.

Cross-Domain Autonomy: Integrating aerial, surface, and underwater robots requires adaptive coordination across different communication layers, power constraints, and environmental dynamics. Seamless sensor fusion and real-time control between heterogeneous platforms remain unresolved challenges.

Semantic Interoperability: When robots from different manufacturers join a mission, they may use incompatible ontologies for describing tasks, environments, and states. Semantic web technologies could enable interoperability but are rarely deployed in robotics systems.

8.6. Deployment and Operational Challenges

Energy and Computation Awareness: Coordinating motion and communication to minimize energy consumption is critical for long-duration missions, particularly in search-and-rescue or planetary exploration settings. Protocol analysis highlights power consumption differences, but dynamic energy-aware task allocation that considers battery states, charging station locations, and mission duration remains underdeveloped. Range-constrained task allocation algorithms provide theoretical guarantees but often assume perfect battery models that fail to capture real degradation patterns.

Long-Duration Autonomy: Agricultural and environmental monitoring missions span days to months. Systems must handle hardware degradation, software bugs, environmental changes (seasonal vegetation growth, weather), and rare failure modes (sensor occlusions, unexpected obstacles). Predictive maintenance, self-diagnostics, and graceful degradation mechanisms are critical but rarely implemented. Continual learning approaches that adapt to evolving environments without catastrophic forgetting remain nascent.

Field-Scale Validation: Many frameworks remain simulation-based or validated only in controlled laboratory settings. Real-world deployments expose unforeseen challenges including GPS errors, radio interference, terrain variability, and unexpected human interactions. Comprehensive field trials are essential to assess true system performance. Even systems validated in photo-realistic simulations encounter sim-to-real gaps when deployed on physical hardware. Standardized outdoor datasets with diverse weather conditions, terrain types, and interference patterns are needed.

Formal Verification at Scale: Temporal logic-based task synthesis provides correctness guarantees but suffers from state space explosion beyond small teams. Compositional

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

synthesis methods that decompose large missions into provably correct sub-tasks are needed to scale formal methods to 50+ robot teams. Current approaches handle individual and collaborative specifications but struggle when team size grows or task complexity increases.

Application-Specific Constraints: Generic coordination algorithms often fail to meet domain-specific requirements (e.g., narrow aisle navigation in warehouses, water sample collection timing in environmental monitoring, off-road traversability uncertainty). Developing domain adaptation frameworks that efficiently tune coordination strategies to application constraints is critical for broader adoption. Transfer learning from simulation or one domain to another remains challenging when environmental statistics differ significantly.

Long-Duration Underground Operations: LAMP 2.0 demonstrates resilience in 6 km underground deployments, but missions extending to days or weeks (e.g., mine rescue, planetary caves) introduce new failure modes. LiDAR sensors accumulate dust and debris degrading scan quality, wheel odometry drifts due to slippage on muddy terrain, and batteries degrade reducing operational endurance. Predictive maintenance models that anticipate sensor degradation and trigger recalibration or robot replacement before catastrophic failure are critical but rarely implemented. Continual SLAM systems that incrementally refine maps while pruning outdated information could handle environment changes (e.g., mine shaft collapses, rock slides), but risk catastrophic forgetting if not carefully designed.

RL Policy Robustness and Sim-to-Real Gap: While RL-based task allocation demonstrates impressive simulation performance, real-world validation reveals brittleness to distributional shifts. Objects heavier than training distribution ranges, novel obstacle geometries not seen in simulation, and unexpected robot failures (e.g., gripper malfunction) cause policy degradation. Domain randomization improves robustness but cannot cover all real-world corner cases. Meta-reinforcement learning that rapidly adapts to deployment-time distribution shifts, or hybrid systems combining RL policies with symbolic planners for fallback behavior, could bridge the gap. However, safety-critical applications may still require formal verification of RL policies under bounded perturbations, an area where tools remain immature.

9. Future Research Directions

Building on recent advances in communication-efficient learning, connectivity-aware coordination, multi-modal perception, and human-robot interaction, several promising directions can further strengthen the scalability, robustness, and real-world applicability of multi-robot systems. We organize these directions thematically to highlight connections between communication infrastructure, coordination algorithms, perception pipelines, and deployment considerations. Table 5 provides a systematic mapping between open challenges identified in Section 8 and corresponding future research directions.

9.1. Communication and Networking Directions

Adaptive Hybrid Protocol Stacks: Extend protocol performance analysis to runtime protocol switching. Robots should autonomously negotiate protocol transitions based on task phase (exploration using LoRa for long-range communication, clustering with Zigbee for mesh networking, manipulation using Bluetooth for low-latency interaction) and environmental conditions (interference levels, distance changes, obstacle density).

Concurrent Transmission for Heterogeneous Platforms: Building on split architecture principles, future work should develop adaptive synchronization schemes that accommodate heterogeneous processing speeds. This would allow slower robots (e.g., resource-constrained micro-drones) to participate in concurrent transmission networks without degrading overall network performance or requiring hardware upgrades.

Semantic Communication Networks: Move beyond bit-level transmission to task-relevant information exchange. Instead of transmitting raw sensor data, robots could share semantic scene graphs, detected anomalies, or task-relevant features, drastically reducing bandwidth while preserving decision-making quality in collaborative perception scenarios.

Resilient Wireless Networking: Designing self-healing, adaptive mesh networks capable of reconfiguration after link failures will enhance communication reliability in dynamic environments. Future systems should jointly optimize task trajectories and connectivity maintenance, integrating predictive learning models with online corrections to prevent communication dropouts during exploration.

Communication-Aware Exploration and Planning: Future systems can integrate predictive communication models that estimate signal strength and dynamically deploy relay nodes to maintain connectivity in complex or subterranean environments. Reinforcement learning can be applied to optimize where and when to deploy mobile relay nodes, minimizing latency and ensuring consistent data flow.

Lightweight Privacy-Preserving Coordination: Extend homomorphic encryption methods with domain-specific optimizations (e.g., trajectory quantization, approximate encryption for non-critical data) to reduce computational overhead. Explore differential privacy and secure multi-party computation as alternatives to FHE for specific coordination tasks where full encryption is unnecessary.

Byzantine-Resilient Coordination: Develop distributed consensus algorithms that maintain performance with up to 40-50% Byzantine agents through redundancy, reputation systems, and outlier detection. Integrate cryptographic verification with statistical anomaly detection for robust multi-robot systems in adversarial environments.

9.2. Distributed SLAM and Mapping Directions

Learned Loop Closure Prioritization: While Swarm-SLAM's spectral method provides principled loop closure selection, it relies on graph-theoretic metrics that may not capture task-specific information value. Future work could

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Table 5
Mapping of Open Challenges to Future Research Directions

Challenge Area	Current Limitations	Future Research Directions	Expected Impact
Scalable Communication	Bandwidth bottleneck beyond 100 robots; network partitioning in large teams	Adaptive hybrid protocol stacks; semantic communication networks; hierarchical information exchange	Enable 1000+ robot coordination
Intermittent Connectivity	Information staleness; conflict resolution upon reconnection; gossip overhead	Communication-aware exploration; predictive relay deployment; lightweight eventually-consistent protocols	Robust operation in GPS-denied and underground environments
Protocol Adaptability	Static protocol selection; no runtime switching; environment-agnostic	Runtime protocol negotiation; task-phase-aware switching; cross-layer optimization	30-50% improvement in efficiency
Privacy and Security	Homomorphic encryption 10-100× overhead; Byzantine tolerance <30%	Lightweight cryptographic methods; differential privacy; Byzantine-resilient consensus (40-50% tolerance)	Secure coordination in adversarial settings
Distributed SLAM Scalability	Quadratic loop closure verification; convergence issues under partition	Learned loop closure prioritization; hierarchical SLAM; semantic-geometric fusion	Scale to 100+ robots over 50+ km
Human-Swarm Interaction	Limited to 2-3 operators; attention overload; trust calibration	AR/VR interfaces; adaptive autonomy; intent prediction; automated summarization	Single operator managing 50+ robots
Sim-to-Real Transfer	RL policy brittleness; distribution shift; sensor noise sensitivity	Domain randomization; meta-RL; hybrid symbolic-neural systems; formal verification	Reliable real-world deployment
Long-Duration Autonomy	Hardware degradation; catastrophic forgetting; rare failure modes	Predictive maintenance; continual learning; graceful degradation mechanisms	Week-to-month autonomous missions
Energy Efficiency	Static power management; no joint motion-communication optimization	Energy-aware task allocation; adaptive duty cycling; wireless power transfer coordination	2-3× mission duration extension
Formal Verification	State explosion beyond 10 robots; limited to simple specifications	Compositional synthesis; probabilistic temporal logic; assume-guarantee reasoning	Provably correct 50+ robot coordination
Heterogeneous Coordination	Incompatible ontologies; sensor fusion challenges; varying capabilities	Semantic interoperability standards; cross-platform middleware; capability-aware allocation	Seamless multi-vendor robot teams
Field Validation	Simulation-only evaluation; controlled lab conditions; missing datasets	Standardized benchmarks; hardware-in-the-loop testbeds; cross-domain transfer datasets	Accelerated real-world deployment

train neural networks to predict loop closure utility based on mission objectives. For example, exploration tasks might prioritize loop closures that reduce frontier uncertainty, while inspection tasks might prioritize revisits to regions of interest. Meta-learning approaches could enable quick adaptation of prioritization strategies to new mission types.

Semantic-Geometric Fusion for Robust Place Recognition: Combining SlideSLAM's object-level semantics with geometric features could provide complementary robustness: semantics handle viewpoint changes, while geometry disambiguates repeated objects through fine-grained spatial structure. Attention-based fusion mechanisms could dynamically weight semantic vs. geometric cues based on environment context (semantic-dominant in structured indoor spaces, geometry-dominant in natural outdoor settings). End-to-end learning of these fusion policies from multi-modal sensor data could discover optimal strategies beyond handcrafted rules.

Hierarchical SLAM for Extreme-Scale Environments: LAMP 2.0 demonstrates scalability to 6 km underground,

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

but planetary cave exploration or urban search-and-rescue may span 50-100 km with hundreds of robots. Hierarchical SLAM architectures that maintain coarse global topology at low resolution and fine-grained local maps at high resolution could enable such scaling. Robots would exchange topological summaries (graph nodes representing regions, edges representing connectivity) rather than full pose graphs, with detailed maps transmitted only when robots operate in the same region. Developing efficient hierarchical representations and ensuring consistency across abstraction levels are key challenges.

Lifelong SLAM with Continual Learning: Long-duration missions require robots to adapt to environmental changes (seasonal vegetation growth, construction activity) without catastrophic forgetting of previously mapped regions. Continual learning frameworks that incrementally incorporate new observations while preserving critical past information through memory replay or elastic weight consolidation could enable persistent SLAM. Combining these with change detection algorithms that identify stable vs. dynamic map components would improve robustness in non-stationary environments.

Extending Semantic-Relational SLAM to Unstructured Environments: Multi S-Graphs [21] demonstrates that room-wall-floor hierarchical representations enable highly efficient collaborative SLAM in structured buildings. Extending these principles to unstructured environments requires developing alternative semantic hierarchies: for outdoor settings, terrain type regions (forest, clearing, rocky terrain), topographic features (ridges, valleys), and landmark clusters could provide analogous structural abstractions. For underwater environments, geological features (reef formations, sediment types) could serve as hierarchical elements. Learning environment-adaptive abstraction hierarchies from data, rather than relying on predefined structural templates, could generalize situational graph SLAM across diverse deployment domains.

9.3. Active SLAM and Exploration Directions

Multi-Objective Active SLAM: Extend frontier selection beyond information gain to jointly optimize coverage, energy consumption, communication maintenance, and risk avoidance. Multi-objective evolutionary algorithms or scalarization methods with learned weights could generate diverse exploration strategies along the Pareto frontier, enabling human operators to select behavior appropriate for mission context (e.g., prioritize safety in search-and-rescue, prioritize speed in time-critical reconnaissance).

Communication-Aware Exploration Planning: Current Active SLAM methods largely ignore communication constraints, assuming robots can always transmit map updates. Future work should integrate connectivity prediction models that forecast signal strength along candidate trajectories, biasing exploration toward paths that maintain network connectivity. When connectivity cannot be preserved, systems should preemptively plan rendezvous points where robots periodically meet to synchronize, balancing exploration progress against synchronization frequency.

Compression-Integrated Exploration: Task-driven map compression [42] demonstrates that significant bandwidth savings are achievable through intelligent abstraction template selection, but current compression methods operate independently from exploration planning. Future work should close this loop by jointly optimizing what information to collect (exploration), how to compress it (abstraction template selection), and when to transmit (communication scheduling). Robots could learn compression-aware exploration policies that preferentially visit regions whose map structure is amenable to high compression ratios, maximizing the information value of each transmitted byte.

Semantic-Guided Active SLAM: Integrate vision-language models to guide exploration toward mission-relevant regions. Rather than uniform frontier selection, robots could prioritize exploring areas predicted to contain task-relevant objects or scene types (e.g., "search for survivors likely in residential structures" biases exploration toward building interiors). Combining VLM scene classification with occupancy-based frontier detection could significantly improve search efficiency in large environments.

9.4. Reinforcement Learning and Adaptive Allocation Directions

Sim-to-Real Transfer for Task Allocation: Current RL policies degrade under real-world distribution shifts. Develop domain adaptation techniques that quickly fine-tune policies using small amounts of real-world data, or hybrid systems combining RL for nominal behavior with rule-based fallbacks for out-of-distribution states. Adversarial training where simulation includes intentionally perturbed dynamics and sensor noise could improve robustness, as could model-based RL that explicitly learns environment dynamics and plans using learned models.

Multi-Agent RL for Decentralized Allocation: Two-stage RL demonstrates hierarchical decomposition benefits but remains centralized. Future work could develop fully decentralized multi-agent RL where each robot independently learns allocation policies through local observations and neighborhood communication. Graph neural networks could enable agents to reason about team-wide state through message passing, avoiding centralized bottlenecks while maintaining near-optimal allocations.

GNN-Based Assignment with Dynamic Re-Allocation: DGNN-GA [22] demonstrates that GNNs can efficiently solve static goal assignment in a decentralized manner, but real-world missions require dynamic re-assignment as goals are completed, new tasks arrive, or robots fail. Extending GNN-based assignment to online settings where the heterogeneous graph evolves over time requires recurrent or attention-based architectures that maintain assignment history and anticipate future task arrivals. Additionally, handling unequal robot-to-goal ratios (more goals than robots or vice versa) and incorporating robot capability constraints into the graph structure remain important extensions.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Adaptive Heterogeneity in Multi-Agent RL: HetGPPO [6] identifies a fundamental trade-off between policy heterogeneity (which improves real-world robustness) and parameter sharing (which improves sample efficiency). Future work should develop mechanisms that dynamically tune the degree of heterogeneity during training: agents could begin with fully shared parameters for fast initial learning and gradually specialize as training progresses, with the divergence rate controlled by task complexity and environmental noise levels. Curriculum-based approaches that incrementally introduce heterogeneity challenges (first physical heterogeneity, then behavioral) could further improve training stability.

Continual RL for Non-Stationary Environments: Warehouse layouts change, new product SKUs arrive, and robot capabilities evolve through hardware upgrades. RL policies trained on static environments fail when distributions shift. Continual RL frameworks that incrementally adapt policies while preventing catastrophic forgetting of previous tasks are essential. Meta-learning could enable rapid adaptation to novel tasks (e.g., new object types) using few-shot examples, while memory-augmented architectures could explicitly store and retrieve task-specific strategies.

Safe RL with Formal Verification: Safety-critical applications require guarantees that RL policies will not enter unsafe states (e.g., collisions, deadlocks). Shield synthesis methods that compute maximal safe subsets of the state-action space, or runtime monitors that verify policy outputs before execution, could provide these guarantees. Combining RL for performance with control barrier functions for safety offers a promising direction, but efficiently computing CBFs for high-dimensional systems remains challenging.

9.5. Human-Robot Interaction Directions

Scalable Multi-Human Multi-Robot Interfaces: Extend interface design principles to hierarchical operator teams where field supervisors manage local robot clusters while central commanders coordinate global strategy. Research attention management algorithms that automatically surface critical alerts to appropriate operators based on task priority, operator expertise, and cognitive load.

Explainable and Trustworthy LLM Planners: Large Language Model-based planners require transparency and reliability. Future research should include reasoning trace visualization, safety filters, and human-in-the-loop validation mechanisms. Runtime explanation generation should proactively clarify robot decision-making when deviations from expected behavior occur.

Human-Centric Semantic Collaboration: Future work should focus on grounding semantic understanding into executable low-level control policies, enabling robots to interpret high-level human intent expressed through natural language, gestures, or context. Integrating multi-turn dialogue systems with adaptive task planning will enhance mutual understanding between humans and robot teams, enabling robots to reason about goals, constraints, and social context dynamically.

Intent Recognition Through Physical Interaction: Develop learning-based intent classifiers that infer human goals from force patterns during co-manipulation. This enables implicit communication in noisy environments where verbal commands fail, extending mechanical communication principles to more sophisticated human-robot collaboration scenarios.

Human-Swarm Interaction: Future systems should include intuitive AR/VR and speech-based interfaces to allow human operators to supervise and collaborate with large robot fleets efficiently, reducing cognitive burden while maintaining situational awareness across distributed teams.

9.6. Perception and Learning Directions

Federated Collaborative Perception: Generalize centralized collaborative perception to federated architectures where robots train local perception models (object detectors, semantic segmentors) and share model updates rather than raw images or centralized features. This preserves privacy while maintaining performance and reduces communication overhead in bandwidth-constrained scenarios.

Multi-Modal Collaborative Perception: Fusing RGB, thermal, LiDAR, and radar modalities can improve situational awareness under poor visibility and GPS-denied conditions, supporting robust distributed mapping and object recognition across diverse environmental conditions.

Continual Learning for Long-Duration Missions: Robots operating for extended periods must incrementally learn new object categories (e.g., emerging product SKUs in warehouses, new pest species in agricultural fields) without catastrophic forgetting. Techniques like elastic weight consolidation, memory replay, and dynamic architectures are essential for maintaining performance over mission lifetimes.

Multi-UAV Occupancy Prediction at Scale: Extend occupancy prediction frameworks to dynamic environments with moving obstacles (vehicles, pedestrians) and integrate temporal prediction. Recurrent neural networks or transformers over time-series occupancy could enable proactive collision avoidance in complex aerial environments.

Exploration with Semantic-Geometric Fusion: Combine geometric Voronoi partitioning approaches with vision-language model-based semantic reasoning. Robots could prioritize exploring regions likely to contain mission-relevant objects (e.g., "search for survivors in areas resembling residential structures"), improving search efficiency in disaster response scenarios. Neural bipartite matching demonstrates the potential of learned frontier assignment but requires extension to semantic priors.

Asynchronous Federated and Decentralized Learning: Distributed learning frameworks can be extended to handle asynchronous communication, packet delays, and intermittent participation among robots, ensuring stable convergence in large-scale deployments with unreliable network connectivity.

Meta-Learning for Communication Strategies: Develop meta-reinforcement learning frameworks that enable robots to quickly adapt learned communication policies to

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

new team compositions, environments, and tasks. Focus on few-shot adaptation using small amounts of real-world data to bridge the sim-to-real gap. Current systems that learn pose error correction through communication could benefit from meta-learning to generalize across sensing modalities and robot platforms.

Hybrid Localization Systems: Combine range measurements, IMU, and opportunistic visual features in unified estimation frameworks that gracefully degrade under sensor failures. Explore event-triggered localization where robots actively seek informative measurements when uncertainty exceeds thresholds. Extend distributed graph optimization approaches to handle dynamic network topologies and asynchronous updates more efficiently.

9.7. Coordination and Control Directions

Energy- and Bandwidth-Aware Mission Planning: Co-optimizing motion trajectories, communication frequency, and computational load can extend operational endurance for field missions involving power-limited robot teams. Develop auction-based task allocation algorithms that incorporate battery state, distance to charging stations, and task urgency, enabling robots near depletion to gracefully hand off tasks before failure.

Collaborative Mapping Under Uncertainty: Enhancing decentralized SLAM algorithms to account for moving objects, uncertain observations, and intermittent connectivity can yield more accurate and resilient maps. Moving beyond centralized semantic map fusion will eliminate bottlenecks and vulnerability to server failures while addressing the challenge of assigning non-overlapping goals as environments and team sizes scale. Asynchronous pose graph optimization offers a path forward but requires integration with active exploration strategies.

Hybrid Physical-Semantic Communication: Future research should explore how force cues, haptic feedback, and physical coupling can be fused with symbolic or language-based coordination, leading to more intuitive and resilient multi-robot teamwork in close human proximity. Incorporating communication-awareness into collaborative mapping and exploration can enable robots to proactively reroute around weak-signal regions while maintaining coverage efficiency.

Hierarchical Macro-Action Policies: Extend macro-action learning to multi-level hierarchies where robots coordinate on strategic (hours), tactical (minutes), and reactive (seconds) timescales. Investigate abstraction learning that automatically discovers useful macro-actions from primitive experiences. Current centralized Q-learning approaches could benefit from hierarchical decomposition to scale beyond warehouse-sized domains.

Game-Theoretic Multi-Robot Control: Generalize game-theoretic CBF methods to teams with heterogeneous dynamics, objectives, and constraints. Develop computational methods for Nash equilibrium selection when multiple equilibria exist, considering fairness, efficiency, and robustness criteria. Extend social navigation frameworks to outdoor and unstructured environments where geometric constraints are less well-defined.

9.8. Cross-Domain and Application-Specific Directions

Cross-Domain and Multi-Platform Autonomy: Extending collaboration to heterogeneous teams of aerial, ground, and marine robots can enable multi-domain missions such as coastal monitoring, search and rescue, and distributed sensing. Adaptive coordination across different communication layers, power constraints, and environmental dynamics remains critical. Water sampling missions demonstrate heterogeneous coordination benefits but require further work on dynamic task reallocation under changing environmental conditions.

Agriculture-Specific Protocol Optimization: Develop specialized agricultural communication standards that integrate protocol selection with agronomic parameters (crop type, growth stage, field topology). For example, vineyards with dense foliage may require more relay nodes and mesh networking (Zigbee) than open wheat fields where long-range protocols (LoRa) suffice.

Warehouse-Manufacturing Convergence: Extend logistics frameworks to incorporate manipulation tasks (picking, packing, assembly). Integrate visual servoing, force control, and multi-modal fusion for tighter human-robot collaboration in assembly lines, bridging the gap between mobile robots and stationary manipulators. Narrow-aisle navigation techniques could be extended to include manipulation constraints.

Domain Adaptation for Coordination: Develop transfer learning frameworks that adapt coordination policies from simulation or one application domain to another (e.g., warehouse navigation to hospital logistics). Focus on identifying domain-invariant coordination principles that transfer across environments. Risk-aware navigation methods could inform domain adaptation by characterizing terrain uncertainty in new environments.

Formal Methods for Large-Scale Systems: Investigate compositional synthesis and assume-guarantee reasoning to scale temporal logic planning to 50+ robot teams. Explore probabilistic temporal logic that relaxes hard constraints to soft preferences, reducing state space while maintaining mission success guarantees. Current temporal logic planners handle individual and collaborative specifications but need hierarchical decomposition for scalability.

Underground-to-Underwater Transfer: LAMP 2.0's success in underground environments suggests applicability to underwater robotics, which faces similar challenges (GPS denial, communication constraints, perceptual degradation from turbidity). Adapting spectral loop closure prioritization and robust optimization to acoustic sensing modalities (sonar, imaging sonar) could enable multi-Autonomous Underwater Vehicle (AUV) collaborative mapping for oceanographic research, pipeline inspection, and search-and-recovery missions. However, acoustic propagation differs fundamentally from RF, requiring protocol redesign and synchronization

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

mechanisms tolerant of multi-second communication latencies.

Opportunistic Space Exploration: MOCHA's intermittent connectivity tolerance aligns well with planetary exploration scenarios where rovers and aerial drones experience communication blackouts due to terrain occlusion or orbital relay limitations. Extending MOCHA to handle multi-hour disconnections (Mars rovers operating autonomously during communication windows when Earth is below the horizon) requires sophisticated information prioritization and conflict resolution upon reconnection. Developing mission-critical message identification and guaranteed eventual delivery protocols could enable resilient multi-robot exploration on planetary surfaces.

9.9. Validation and Standardization Directions

Field-Scale Hardware Validation: Many existing frameworks remain simulation-based. Real-world deployments are essential to assess communication latency, bandwidth contention, and energy consumption in practical settings. Establishing shared physical testbeds and datasets with real-world sensor traces would accelerate community progress. Photo-realistic simulations provide intermediate validation but cannot capture all real-world phenomena.

Standardized ROS2-Based Frameworks: Extending ROS2 with built-in communication monitoring, distributed learning modules, and middleware support can simplify large-scale heterogeneous robot deployment in real-world environments. Standardized interfaces would improve interoperability across platforms and reduce integration overhead.

Cross-Domain Transfer Benchmarks: Create datasets spanning agriculture, warehouses, disaster zones, and other domains. Evaluate whether algorithms trained in one domain generalize to others, identifying key bottlenecks in transfer learning and domain adaptation for multi-robot systems.

Benchmarks for Specialized Domains: Establish standardized benchmarks for narrow warehouse navigation, off-road multi-robot exploration, and water quality monitoring with publicly available datasets, simulators, and evaluation metrics. Include diverse scenarios (nominal, degraded sensors, adversarial conditions) to comprehensively assess robustness. Dense reconstruction and coverage planning benchmarks would also benefit from standardization.

Hardware-in-the-Loop Testing: Develop hybrid simulation-hardware testbeds where some robots are physical and others simulated, enabling large-scale experiments without full hardware deployment costs. Focus on accurate communication and dynamics modeling to ensure sim-to-real transfer.

10. Conclusion

This survey consolidates state-of-the-art research on multi-robot collaboration and communication, emphasizing the paradigm shift from classical centralized architectures toward decentralized, learning-based, and language-driven systems. Communication paradigms now encompass concurrent transmission, protocol-aware selection, semantic intent sharing, and mechanical force-based coordination, enabling collaboration that is increasingly diverse, efficient, and context-adaptive.

The integration of Large Language Models, Vision-Language Models, and Graph Neural Networks has transformed coordination from rigid rule-based systems into interpretable, human-aligned frameworks capable of reasoning about tasks, adapting to novel environments, and incorporating human feedback. Concurrently, distributed learning frameworks and sparse communication strategies enable large-scale teams to share knowledge without overwhelming network bandwidth. Recent advances in distributed SLAM, opportunistic communication, and reinforcement learning-based task allocation demonstrate that large robot teams can now collaboratively map complex environments, maintain coordination through intermittent connectivity, and handle previously intractable operational constraints in real-world deployments.

Reviewed applications across UAV swarms, warehouse logistics, precision agriculture, environmental monitoring, and disaster response underscore deployment breadth. Each domain exposes unique challenges: energy constraints in agriculture, real-time obstacle avoidance in warehouses, narrow-aisle navigation, GPS-denied localization, off-road traversability uncertainty, and semantic reasoning for household tasks. Recent work demonstrates that effective systems must simultaneously address communication infrastructure, human-system interaction, perception and learning, and physical coordination.

Despite progress, significant challenges remain. Scalable communication for 100+ robots, protocol adaptability, privacy-preserving coordination that is computationally efficient, Byzantine resilience, human trust calibration, sim-to-real transfer, relative localization without infrastructure, formal verification at scale, continual learning, and long-duration autonomy demand continued research. Bridging algorithmic innovation with field deployment through resilient networking, standardized benchmarks, and cross-domain validation is essential.

Future directions span adaptive hybrid protocol stacks, lightweight cryptographic coordination, federated collaborative perception, continual SLAM for non-stationary environments, multi-objective active exploration, sim-to-real transfer for RL policies, meta-learning for communication, energy-aware task allocation, hierarchical macro-action policies, game-theoretic control extensions, domain adaptation frameworks, and compositional formal methods. The convergence of semantic communication, mechanical interaction, and digital networking suggests that next-generation systems will be multi-modal by design, leveraging heterogeneous communication channels to achieve robustness, efficiency, and human compatibility.

As multi-robot systems transition from laboratory demonstrations to real-world deployments in agriculture, disaster response, space exploration, and smart cities, the community must prioritize practical considerations: energy efficiency, operator workload, regulatory compliance, and societal impact

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

alongside algorithmic sophistication. This survey provides a comprehensive snapshot of current capabilities, a taxonomy for organizing rapidly expanding literature, and a roadmap for advancing multi-robot collaboration toward scalable, interpretable, and trustworthy autonomous teamwork.

Acknowledgments

The authors acknowledge the use of artificial intelligence–based tools for grammar checking, spelling correction, and minor language refinement during the preparation of this survey manuscript. All literature selection, critical analysis, synthesis, and technical interpretations were performed by the authors.

CRediT authorship contribution statement

Owais Ahmed: Conceptualization, Methodology, Literature review, Writing – original draft, Visualization. **Adeel Khalid:** Supervision, Writing – review & editing. **Daniel Byers:** Literature review, Writing – review & editing. **M. Hassan Tanveer:** Supervision, Methodology, Writing – review & editing.

References

[1] , 2025. Co-navgpt: Multi-robot cooperative visual semantic navigation using vision-language models. Arxiv preprint .

[2] Agizza, G., Isler, V., 2022. Multi-robot collaborative monocular slam utilizing rendezvous, in: IEEE International Conference on Robotics and Automation (ICRA). doi:10.1109/ICRA46639.2022.9812243.

[3] Bai, R., Zheng, R., Liu, M., Zhang, S., 2023. Multi-robot task planning under individual and collaborative temporal logic specifications. IEEE Robotics and Automation Letters 8, 4850–4857.

[4] Barros, T., Maciel, L., et al., 2025. Multi-robot system for autonomous cooperative counter-uas missions: Design, integration, and field testing, in: IEEE International Conference on Robotics and Automation (ICRA).

[5] Bartolomei, L., Karrer, M., Chli, M., 2020. Multi-robot coordination with agent-server architecture for autonomous navigation in partially unknown environments, in: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 1516–1522. doi:10.1109/IROS45743.2020.9341367.

[6] Bettini, M., Shankar, A., Prorok, A., 2023. Heterogeneous multi-robot reinforcement learning, in: arXiv preprint arXiv:2301.07137.

[7] Bharadwaj, S., Gonabattula, K., Saha, S., Sarkar, C., Raja, R., 2021. Concurrent transmission for multi-robot coordination. arXiv preprint URL: https://arxiv.org/abs/2112.00273. arXiv:2112.00273v1 [cs.RO].

[8] Cai, X., Everett, M., How, J., 2022. Risk-aware off-road navigation via a learned speed distribution map, in: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). doi:10.1109/IROS47612.2022.9982193.

[9] Cao, H., Shreedharan, S., Atanasov, N., 2024. Multi-robot object slam using distributed variational inference. IEEE Robotics and Automation Letters 9, 8722–8729. doi:10.1109/LRA.2024.3451389.

[10] Chandra, R., Zinage, V., Bakolas, E., Biswas, J., 2023. Decentralized multi-robot social navigation in constrained environments via game-theoretic control barrier functions, in: IEEE International Conference on Robotics and Automation (ICRA). doi:10.1109/ICRA48891.2023.10160344.

[11] Chang, Y., Ebadi, K., Denniston, C.E., Ginting, M.F., Rosinol, A., Reinke, A., Palieri, M., Shi, J., Chatterjee, A., Morrell, B., Aghamohammadi, A.a., Carlone, L., 2022. Lamp 2.0: A robust multi-robot slam system for operation in challenging large-scale underground environments. IEEE Robotics and Automation Letters 7, 9175–9182. doi:10.1109/LRA.2022.3191204.

[12] Chatziparaschis, D., Lagoudakis, M.G., Partsinevelos, P., 2020. Aerial and ground robot collaboration for autonomous mapping in search and rescue missions. Drones .

[13] Chen, Y., Arkin, J., Zhang, Y., Roy, N., Fan, C., 2025. Scalable multi-robot collaboration with large language models: Centralized or decentralized systems, in: IEEE International Conference on Robotics and Automation (ICRA).

[14] Dey, E., Hossain, J., Roy, N., Busart, C., 2022. Synchrosim: An integrated co-simulation middleware for heterogeneous multi-robot system, in: 2022 18th International Conference on Distributed Computing in Sensor Systems (DCOSS), IEEE. pp. 334–341.

[15] Dong, Q., Xi, H., Zhang, S., Bi, Q., Li, T., Wang, Z., Zhang, X., 2024. Fast and communication-efficient multi-uav exploration via voronoi partition on dynamic topological graph. Accepted for Publication Received: 5 August 2023 / Accepted: 30 May 2024 / Published online: 1 August 2024.

[16] Dong, S., Fan, Q., Wang, H., Wang, J., Yi, L., Funkhouser, T., Chen, B., Guibas, L., 2021. Multi-robot collaborative dense scene reconstruction. ACM Transactions on Graphics (TOG) 40. doi:10.1145/3450626.3459846.

[17] Elwin, M.L., Strong, B., Freeman, R.A., Lynch, K.M., 2023. Human–multirobot collaborative mobile manipulation: The omnid mocobots. IEEE Robotics and Automation Letters 8, 376–383. URL: https://doi.org/10.1109/LRA.2022.3226366, doi:10.1109/LRA.2022.3226366.

[18] Fang, B., Zhi, W., Hu, G., Rosa, S., Henderson, P., Xiao, H., 2023. Learning to communicate and correct pose errors in multi-agent reinforcement learning, in: Conference on Robot Learning (CoRL).

[19] Farhan, M., Ahmed, M.A., Khedr, A., 2025. Efficient multi-robot active slam. Journal of Intelligent & Robotic Systems 111, 1–19. doi:10.1007/s10846-025-02275-8.

[20] Farooq, H., Imran, M., 2021. Multi-agent deep reinforcement learning based spectrum allocation for d2d underlay communications. IEEE Transactions on Cognitive Communications and Networking 7, 127–138. doi:10.1109/TCCN.2020.3024720.

[21] Fernandez-Cortizas, M., Bavle, H., Perez-Saura, D., Sanchez-Lopez, J.L., Campoy, P., Voos, H., 2024. Multi s-graphs: An efficient distributed semantic-relational collaborative slam, in: arXiv preprint arXiv:2401.05152.

[22] Goarin, M., Loianno, G., 2024. Graph neural network for decentralized multi-robot goal assignment. IEEE Robotics and Automation Letters 9, 4051–4058.

[23] Gusrialdi, A., Hatanaka, T., Fujita, M., 2022. Integrating online learning and connectivity maintenance for communication-aware multi-robot coordination. IEEE Transactions on Robotics .

[24] Hu, K., Zou, L., Chen, Z., Jiang, J., Jiang, F., Tao, X., 2021. Wireless multi-robot collaboration: Communications, perception, control, and planning. IEEE Internet of Things Journal 8, 11984–11997. doi:10.1109/JIOT.2021.3060545.

[25] Huang, J., Huang, Y., Zhao, Y., Meng, W., Xu, J., 2025. Cocol: A communication efficient decentralized collaborative method for multi-robot systems. arXiv preprint URL: https://arxiv.org/abs/2508.20898.

[26] Huo, J., Zhen, R., Zhang, S., Liu, M., 2022. Multi-robot path planning in narrow warehouse environments with fast feasibility heuristics, in: Proceedings of the 41st Chinese Control Conference, pp. 5816–5821.

[27] Krizmancic, M., Arbanas, B., Petrovic, T., Petric, F., Bogdan, S., 2020. Cooperative aerial-ground multi-robot system for automated construction tasks. IEEE Robotics and Automation Letters 5, 798–805. doi:10.1109/LRA.2020.2965855.

[28] Kulathunga, G., Klimchik, A., 2022. Heterogeneous multi-robot system for exploration and strategic water sampling. IEEE Robotics and Automation Letters 7, 9761–9768. doi:10.1109/LRA.2022.3189831.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

[29] Kumar, M., Gordon, N., Bagchi, S., Abbasi, A., Joshi, A., 2023. Communication-efficient multi-robot path planning using fully homomorphic encryption, in: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). doi:10.1109/IROS55552.2023.10341791.

[30] Lajoie, P.Y., Ramtoula, B., Chang, Y., Carlone, L., Beltrame, G., 2024. Swarm-slam: Sparse decentralized collaborative simultaneous localization and mapping framework for multi-robot systems. IEEE Transactions on Robotics 40, 4081–4100. doi:10.1109/TRO.2024.3422003.

[31] Latif, E., Parasuraman, R., 2022. Dgorl: Distributed graph optimization based relative localization of multi-robot systems. arXiv preprint ArXiv:2201.03486.

[32] Li, Z., Wu, W., Wang, Y., Xu, Y., Hunt, W., Stein, S., 2025. Hmcf: A human-in-the-loop multi-robot collaboration framework based on large language models. arXiv preprint URL: https://arxiv.org/abs/2505.00820.

[33] Lin, Z., Chen, W., Jin, X., Yang, Y., Fan, L., Zhang, Y., Zhang, Y., Zhang, Z., 2025. Mcop: Multi-uav collaborative occupancy prediction. arXiv preprint URL: https://arxiv.org/abs/2510.12679. arXiv:2510.12679v2 [cs.CV].

[34] Liu, K., Tang, Z., Wang, D., Wang, Z., Li, X., Zhao, B., 2025a. Coherent: Collaboration of heterogeneous multi-robot system with large language models, in: IEEE International Conference on Robotics and Automation (ICRA).

[35] Liu, X., Su, J., Jiang, X., Zheng, Y., Agia, C., Scaramuzza, D., Ang Jr, M.H., Rus, D., 2025b. Slideslam: Sparse, lightweight, decentralized metric-semantic slam for multi-robot navigation. IEEE Transactions on Robotics 41, 653–672. doi:10.1109/TRO.2025.3629786.

[36] Liu, Y., Gao, W., Wang, Z., Williams, R.K., 2024. Cospam: Multi-robot collaborative simultaneous path planning and semantic mapping. IEEE Robotics and Automation Letters .

[37] Mandi, Z., Jain, S., Song, S., 2025. Roco: Dialectic multi-robot collaboration with large language models, in: IEEE International Conference on Robotics and Automation (ICRA).

[38] Manley, J., Puzzuoli, D., Taylor, C., Stahr, F., Angus, J., 2025. A new approach to multi-domain ocean monitoring: Combining uas with usvs, in: Proceedings of MTS/IEEE OCEANS 2025 Conference, IEEE.

[39] Miller, I.D., Cladera, F., Smith, T.J., Taylor, C.J., Kumar, V., 2024. Enabling large-scale heterogeneous collaboration with opportunistic communications, in: 2024 IEEE International Conference on Robotics and Automation (ICRA), pp. 3758–3765. doi:10.1109/ICRA57147.2024.10611469.

[40] Nair, B.R., 2024. Collaborative perception in multi-robot systems: Case studies in household cleaning and warehouse operations. arXiv preprint URL: https://arxiv.org/abs/2408.14039. arXiv:2408.14039v1 [cs.RO].

[41] Patel, J., Ramaswamy, T., Li, Z., Pinciroli, C., 2021. Direct and indirect communication in multi-human multi-robot interaction. arXiv preprint URL: https://arxiv.org/abs/2102.00672. arXiv:2102.00672v1 [cs.RO].

[42] Psomiadis, E., Pedram, A.R., Maity, D., Tsiotras, P., 2025. Communication-aware iterative map compression for online path-planning, in: arXiv preprint arXiv:2503.10843.

[43] Queralta, J.P., 2025. Collaborative multi-robot search and rescue: Planning, coordination, perception, and active vision. IEEE Access .

[44] Sanghvi, N., Niyogi, R., Tripathi, R., 2025. Performance analysis of communication protocols for multi-robot coordination in agricultural field. International Journal of Pervasive Computing and Communications 21, 232–251. doi:10.1108/IJPCC-12-2024-0449.

[45] da Silva, M.S., Mesbahi, M., Wettergreen, D., Barfoot, T.D., Agha, A.A., 2022. Achord: Communication-aware multi-robot coordination with intermittent connectivity. IEEE Robotics and Automation Letters 7, 11021–11028. doi:10.1109/LRA.2022.3192039.

[46] Soria, X., Dorigo, F., Scaramuzza, D., 2025. Swarm-slam: Decentralized and communication-efficient collaborative slam for multi-robot systems. IEEE Robotics and Automation Letters .

[47] Sung, Y., Tokekar, P., Stone, P., 2022. Multi-robot target tracking with sensing and communication danger zones. IEEE Transactions on Robotics 38, 1734–1753. doi:10.1109/TRO.2021.3128693.

[48] Suzuki, K., Jimbo, T., Fujii, R., Takeshita, K., Matsubara, T., 2024. Reinforcement learning of multi-robot task allocation for multi-object transportation with infeasible tasks. arXiv preprint arXiv:2404.11817 .

[49] Tang, J., Xiong, G., Jing, Y., Huang, Z., Zhang, X., 2022a. Multi-robot task allocation and path planning with maximum range constraints. IEEE Transactions on Automation Science and Engineering doi:10.1109/TASE.2021.3127643.

[50] Tang, J., Xiong, G., Jing, Y., Huang, Z., Zhang, X., 2022b. Optimal multi-robot coverage path planning: Ideal-shaped spanning tree. IEEE Robotics and Automation Letters 7, 7391–7398. doi:10.1109/LRA.2022.3189146.

[51] Tian, Y., Chang, Y., Arias, F.H., Nieto-Granda, C., How, J., Carlone, L., 2022a. Resilient and distributed multi-robot visual slam: Datasets, experiments, and lessons learned, in: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). doi:10.1109/IROS47612.2022.9981647.

[52] Tian, Y., Koppel, A., Bedi, A.S., How, J.P., 2022b. Asynchronous and parallel distributed pose graph optimization. IEEE Robotics and Automation Letters 7, 4348–4355. doi:10.1109/LRA.2022.3145063.

[53] Wang, L., Liu, G., 2024. Research on multi-robot collaborative operation in logistics and warehousing using a3c optimized yolov5-ppo model. Frontiers in Neurorobotics 17, 1329589. doi:10.3389/fnbot.2023.1329589.

[54] Wu, B., Suh, C.S., 2025. State-of-the-art in robot learning for multi-robot collaboration: A comprehensive survey. Arxiv preprint .

[55] Yan, L., Stouraitis, T., Vijayakumar, S., 2025. Decentralized ability-aware adaptive control for multi-robot collaborative manipulation. IEEE Robotics and Automation Letters .

[56] Yang, J., Borovikov, I., Dhand, H., Kunes, R., Zhou, H., Iscen, A., Kew, J., Tessler, C., Moufarek, A., Littman, M.L., Vereshchaka, A., Barraquand, J., Petrenko, A., Biyik, E., Maiti, A., 2023. Learning multi-robot decentralized macro-action-based policies via a centralized q-net, in: IEEE International Conference on Robotics and Automation (ICRA). doi:10.1109/ICRA48891.2023.10161125.

[57] Yang, K.M., Han, J.B., Seo, K.H., 2019. A multi-robot control system based on ros for exploring disaster environment, in: Proceedings of the 7th International Conference on Control, Mechatronics and Automation (ICCMA), pp. 168–173. doi:10.1109/ICCMA46720.2019.8988650.

[58] Ye, K., Dong, S., Fan, Q., Wang, H., Yi, L., Xia, F., Wang, J., Chen, B., 2022. Multi-robot active mapping via neural bipartite graph matching, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 14839–14848. doi:10.1109/CVPR52688.2022.01442.

[59] Ye, Z., Li, D., Li, Y., Tang, S., Zhou, J., You, K., 2024. A two-stage reinforcement learning-based approach for multi-entity task allocation. arXiv preprint arXiv:2407.00496 .

[60] Yu, J., Vincent, J.A., Schwager, M., 2025. Dinno: Distributed neural network optimization for multi-robot collaborative learning. IEEE Robotics and Automation Letters .

[61] Zhang, Z., Wei, P., Chen, X., Chen, J., Chen, J., 2023. Distributed range-based relative localization among multiple robots in dynamic networks. IEEE Robotics and Automation Letters 8, 3450–3457. doi:10.1109/LRA.2023.3269320.

[62] Zhou, L., Tokekar, P., 2021. Graph neural networks for decentralized multi-robot submodular action selection, in: International Symposium on Distributed Autonomous Robotic Systems (DARS), Springer.

[63] Zhou, Y., Xiao, J., Zhou, Y., Loianno, G., 2025. Multi-robot collaborative perception with graph neural networks. IEEE Robotics and Automation Letters .

Owais Ahmed is a Doctorate Student at the College of Engineering and Engineering Technology, Kennesaw State University, Georgia, USA. His research interests include multi-robot systems, communication protocols, distributed coordination, and machine learning for robotics.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272

Adeel Khalid is the Dean of Research at the College of Engineering and Engineering Technology, Kennesaw State University, Georgia, USA. His research interests include robotics, automation, control systems, and intelligent systems design.

Daniel Byers is a Doctorate Student at the College of Engineering and Engineering Technology, Kennesaw State University, Georgia, USA. His research interests include robotic systems, distributed computing, and multi-agent coordination.

M. Hassan Tanveer is an Assistant Professor in the Department of Robotics and Mechatronics Engineering, Kennesaw State University, Georgia, USA. His research interests include mechatronics, control systems, sensor integration, and multi-robot coordination.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7067272